Chiari Malformation and Syringomyelia

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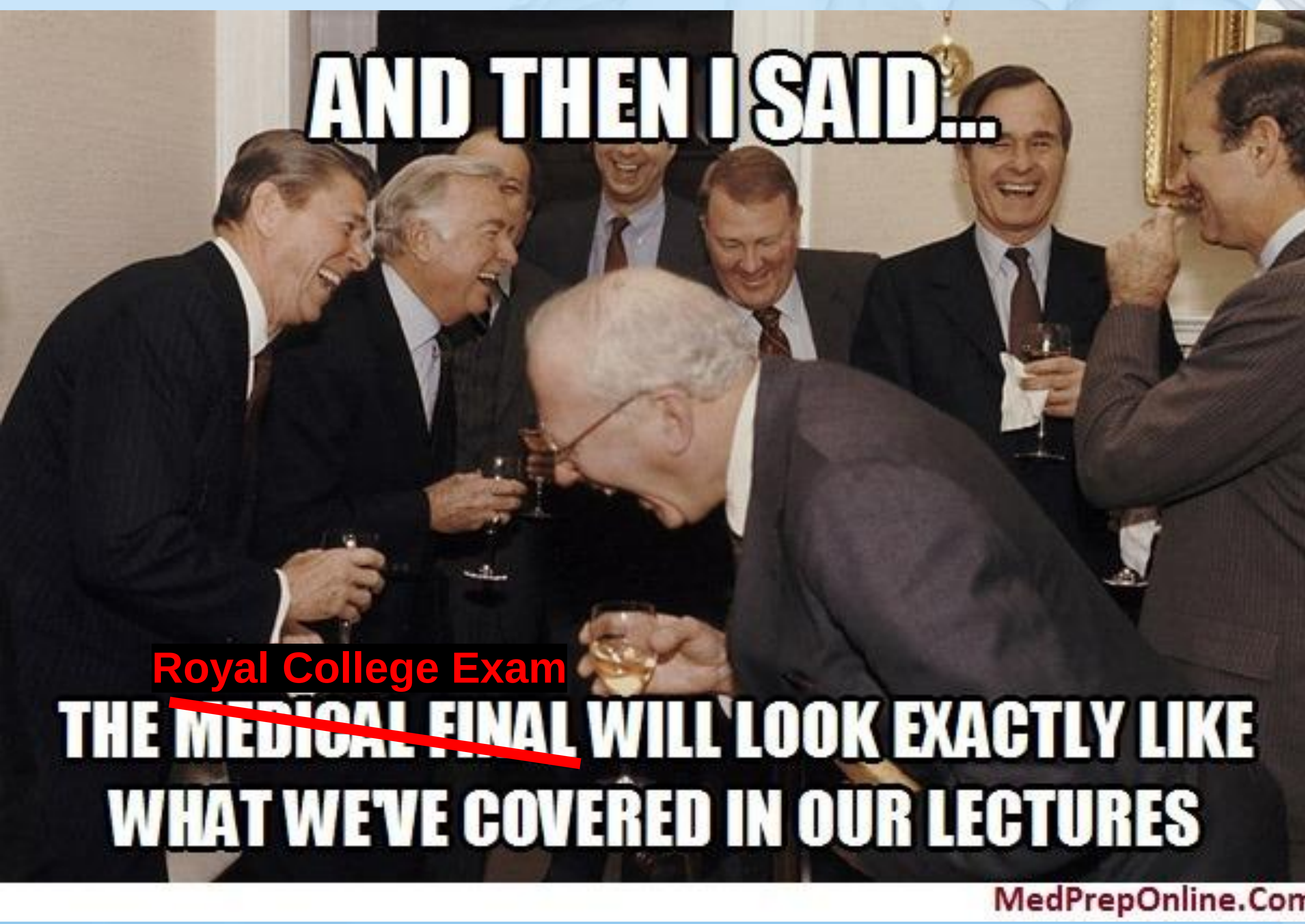
AND THEN I SAID...
Royal College Exam
THE ~~MEDICAL FINAL~~ WILL LOOK EXACTLY LIKE WHAT WE'VE COVERED IN OUR LECTURES
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More Common = More Details
Less Common = Less Details

Hype over the last 5 years

New (but not too new) evidence over the last 5 years

What is exciting over the last 5 years in peds nsx?

Peds oncology – MB/EPN, BRAF, H3K27M, immune checkpoint inhibitors

Peds functional – SDR, DBS, MRgLITT

Peds developmental – Fetal surgery, endoscopic synostosis

Peds trauma – concussion, CTE, neurointensive care

Chiari Malformations

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Disclosures

- None relevant to this talk
- Speaking Fees from LivaNova Inc.
- Investigator Initiated Grant support from LivaNova Inc.

Objectives

1. Describe the definition and classification of “Chiari Malformations”
2. Describe Syringomyelia
3. Explain the association of Chiari I malformation and Syringomyelia and the pathophysiological theories explaining this
4. Choose appropriate therapy of Chiari I malformation with or without syringomyelia

Chiari Classifications

1. Type 1
2. Type 2
3. Type 3
4. Type 4
5. Type 0
6. Type 1.5

Hans Chiari 1851-1916

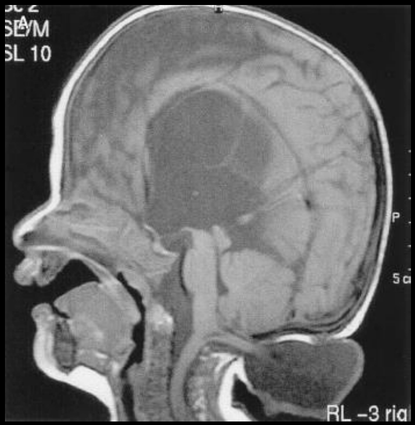
Chiari Classifications

1. Type 1
   - Classically caudal herniation of tonsils >5 mm below FM
   - Not associated with caudal descent of brainstem
   - Hydrocephalus uncommon
2. Type 2
   - Caudal herniation of cervicomedullary junction (vermis, brainstem and 4^{th} ventricle)
   - Almost all have MMC; most hydrocephaluls
   - Diagnosed in infants
3. Type 3
   - Cerebellar herniation into spine canal and FM encephalocele
4. Type 4
   - Cerebellar hypoplasia/Aplasia

Chiari I

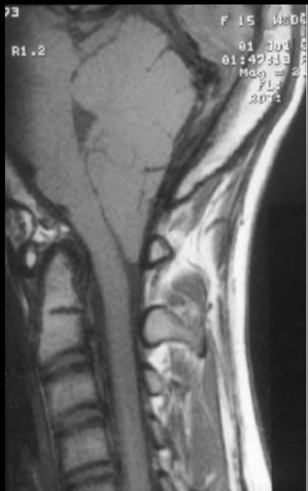
Chiari II

Chiari III



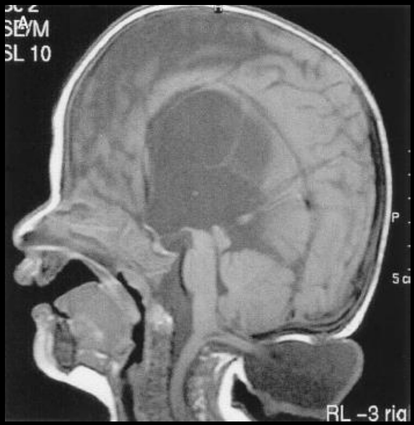
Chiari IV

Chiari I



Chiari II

Chiari III



Chiari IV

Chiari 0

Chiari 1.5

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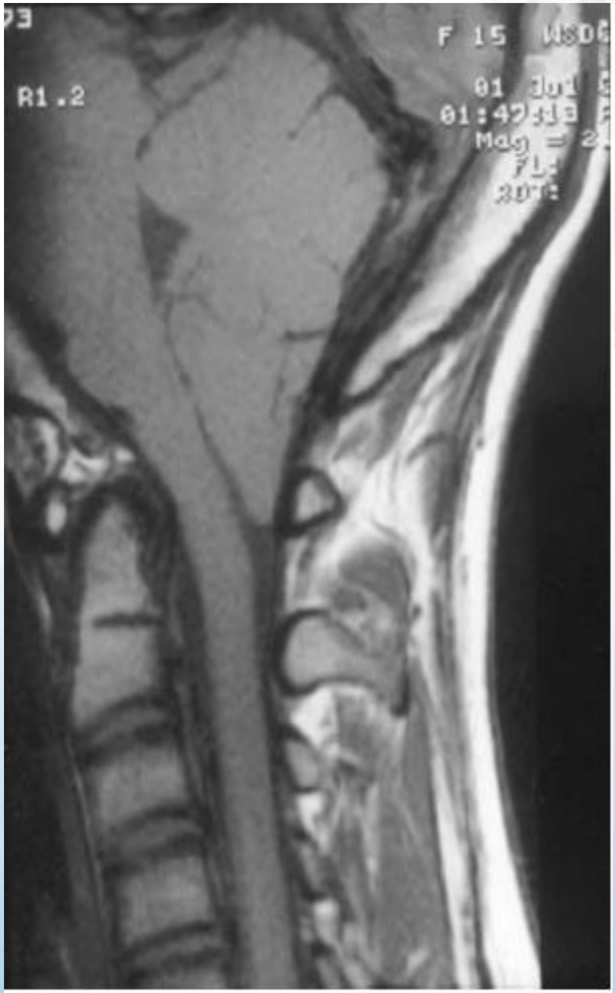
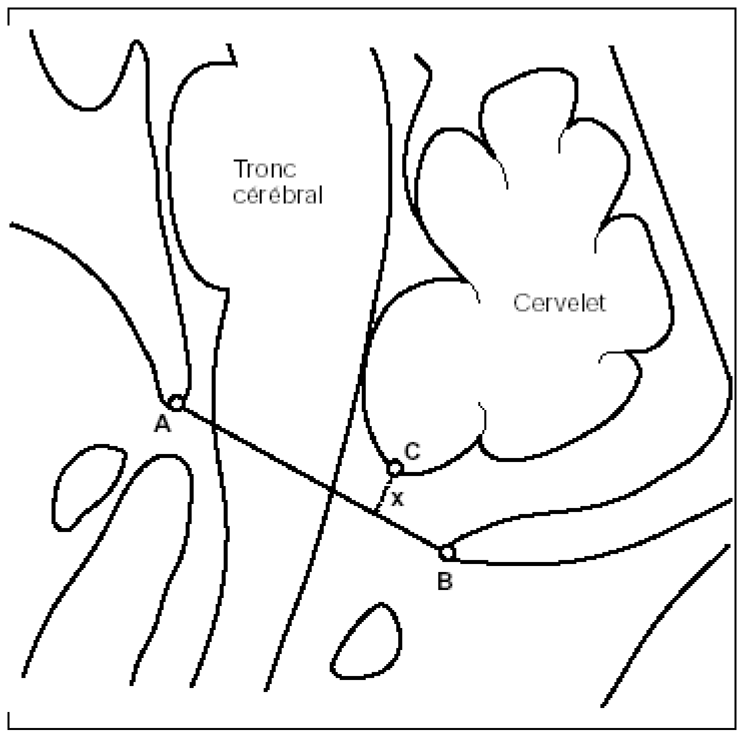
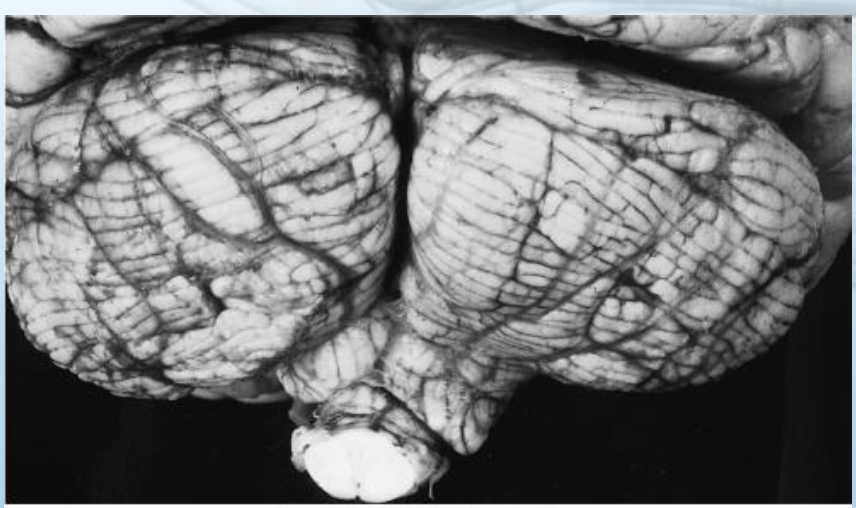
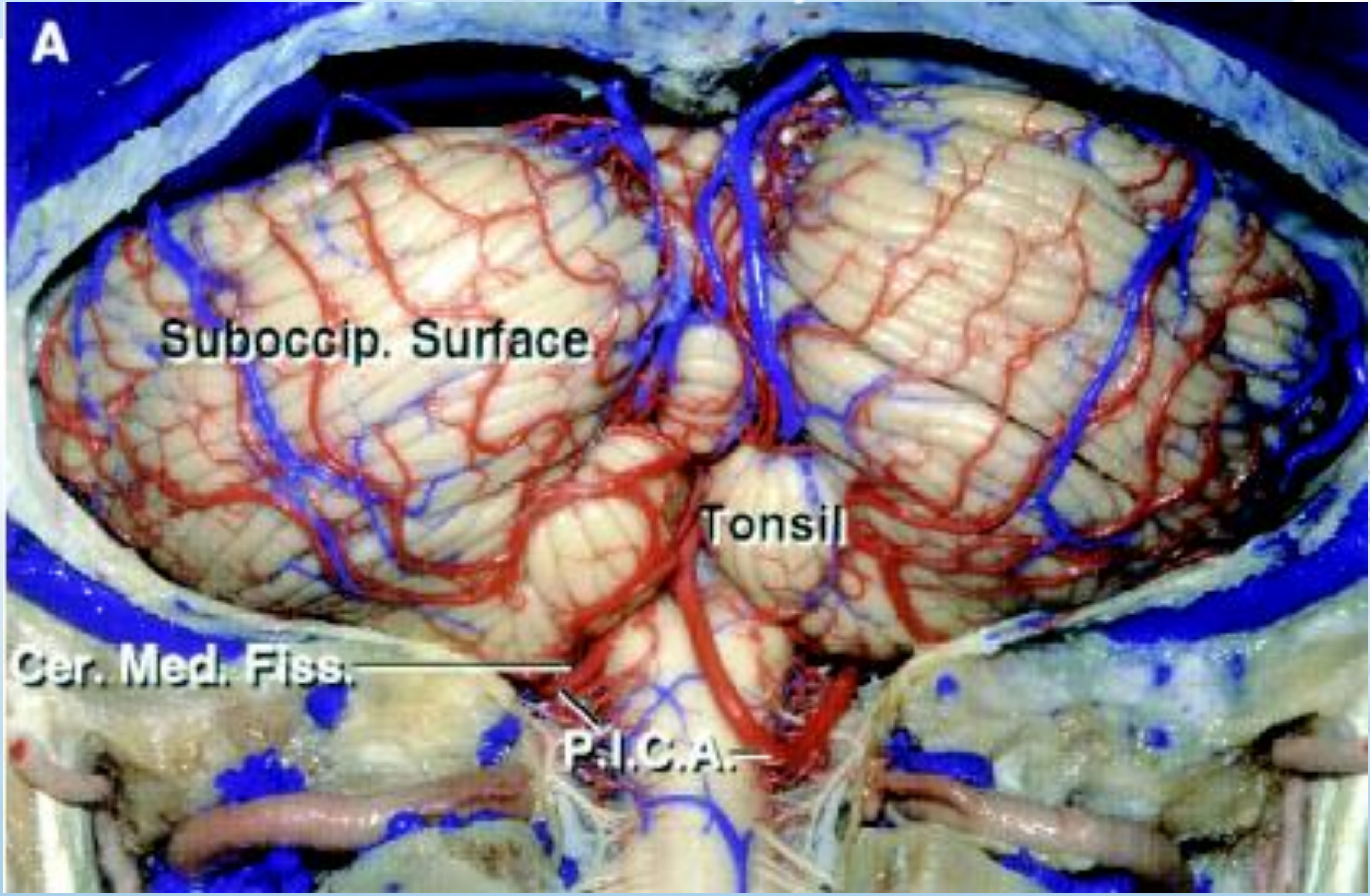
Chiari I Malformation



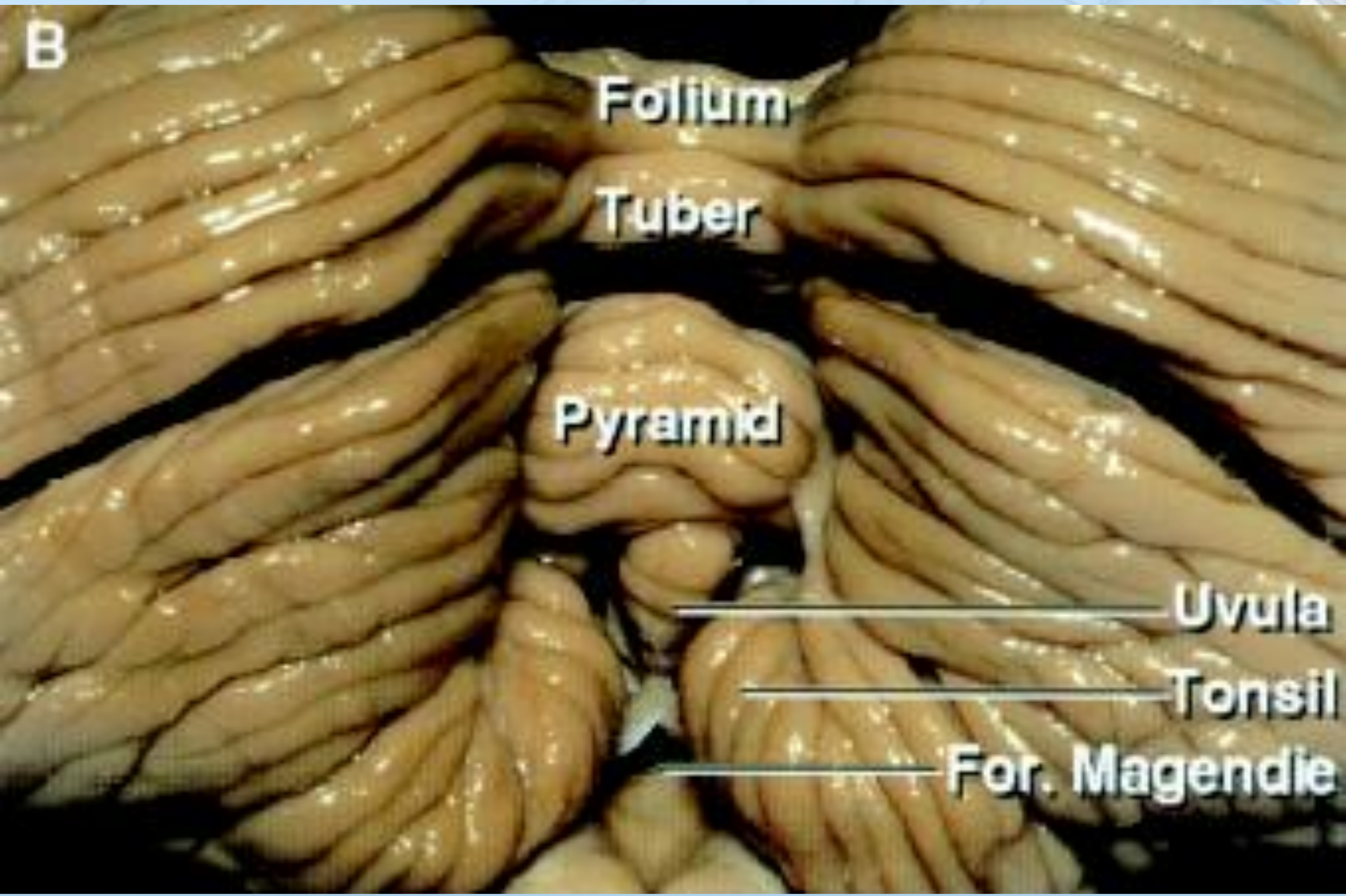




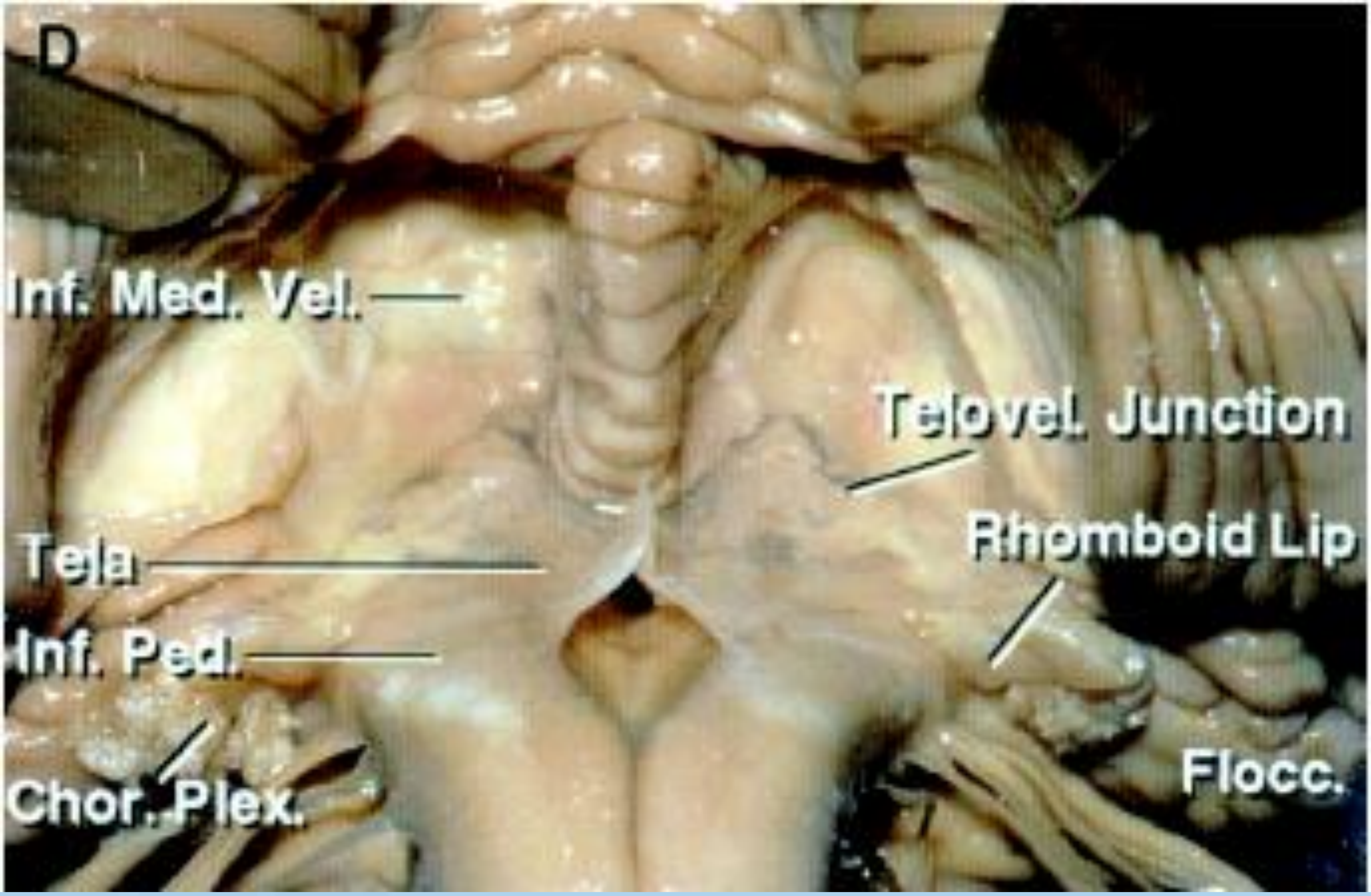
Anatomy



Rhoton, Neurosurgery 2000

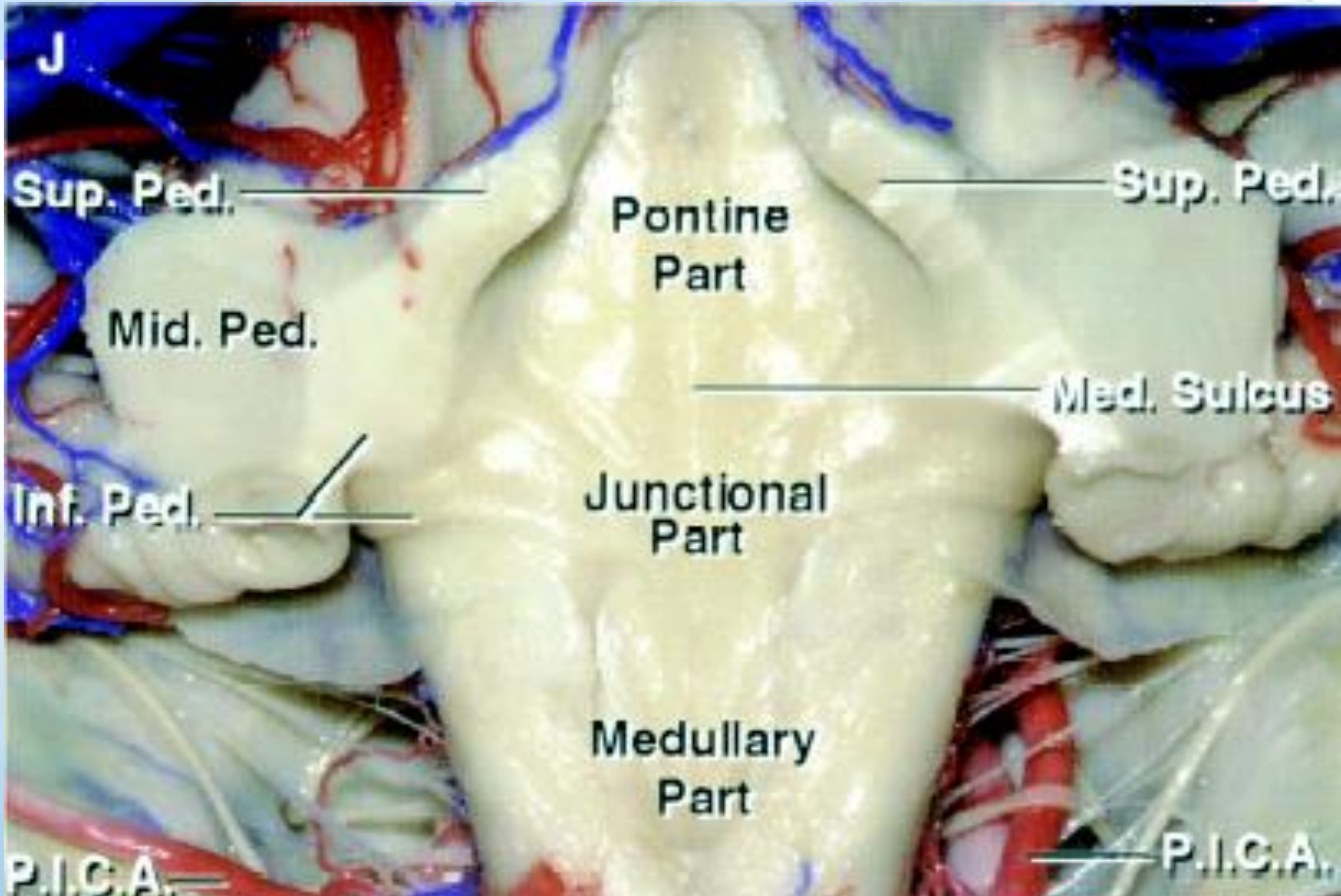


Rhoton, Neurosurgery 2000



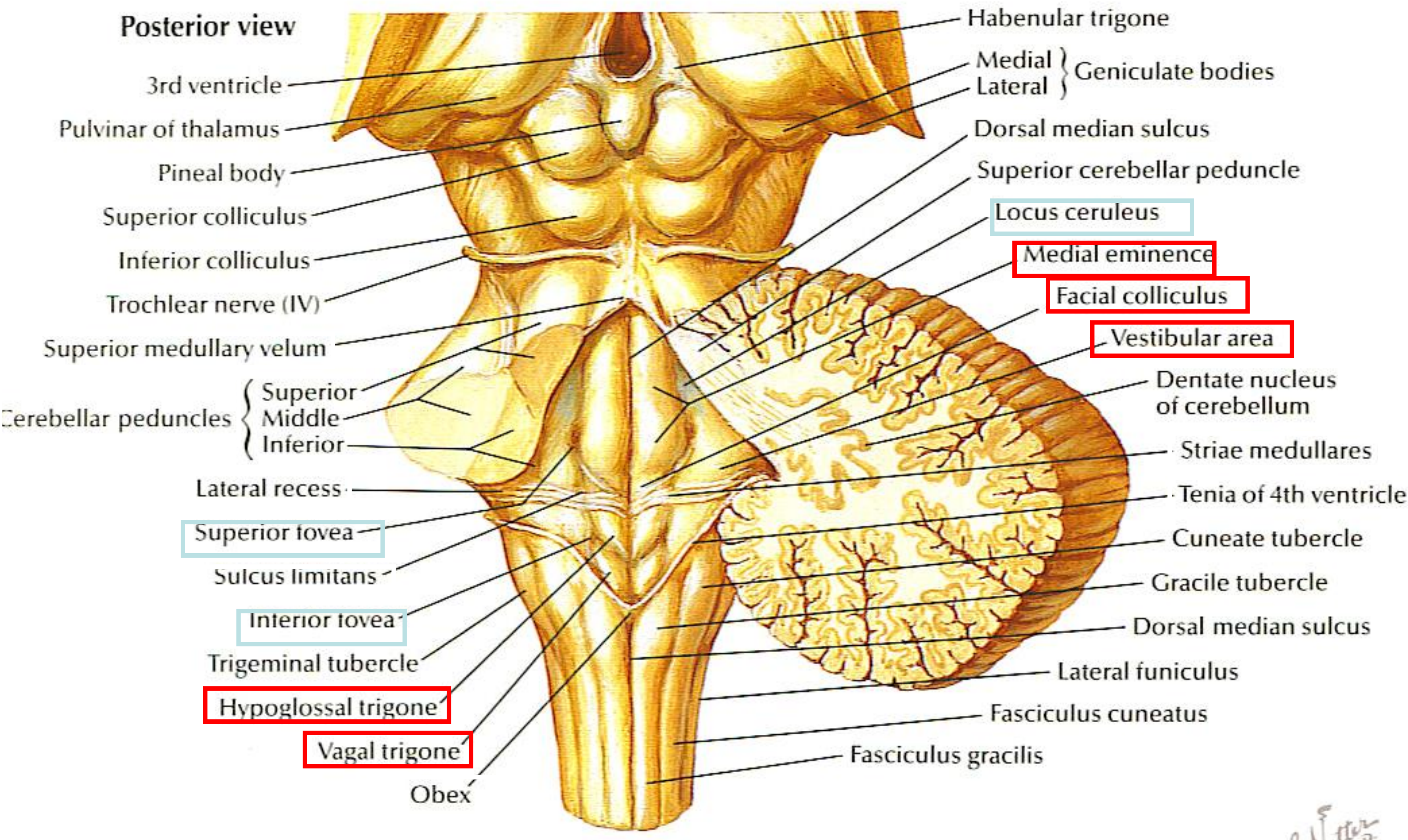
Rhoton, Neurosurgery 2000

Floor of 4th Ventricle



Rhoton, Neurosurgery 2000

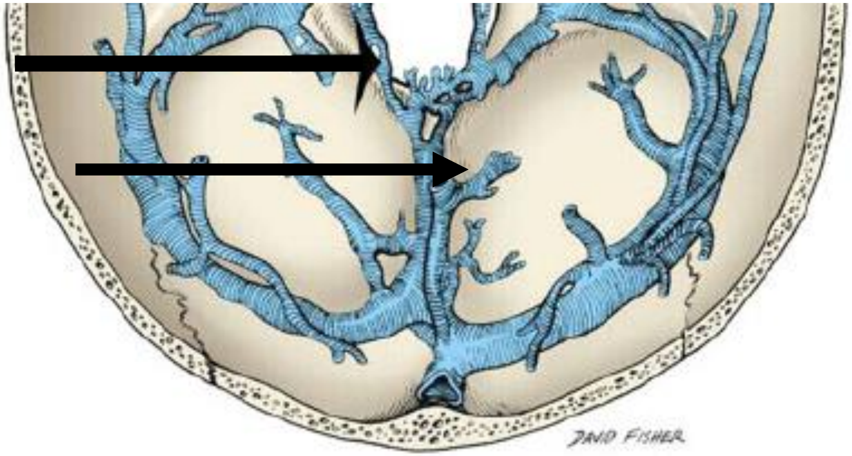
Floor of 4th Ventricle



Venous Sinuses of the FM

Marginal Sinus

Occipital Sinus



Occipital Sinus

The OS was present in 209 of the 555 patients (37.7%)

The OS was observed less frequently in subjects younger than 50 years.

Kobayashi
Neuroradiology 2006

Table 2 Communications of the occipital sinus

Communicating vein	No. of subjects (%)
Cranial	
Confluence of sinuses	25 (11.4%)
Transverse sinus	130 (59.4%)
Straight sinus	10 (4.6%)
Dural vein	58 (26.5%)
Emissary vein	8 (3.7%)
Straight sinus bifurcation	25 (11.4%)
Superior sagittal sinus bifurcation	2 (0.9%)
Superior sagittal sinus	1 (0.5%)
Anastomosis of bilateral TSs	1 (0.5%)
Caudal	
Vertebral venous plexus	97 (44.3%)
Dural vein	116 (53.0%)
Marginal sinus	25 (11.4%)
Sigmoid sinus	26 (11.9%)

Table 1 Frequency of occipital sinus: age- and sex- differences

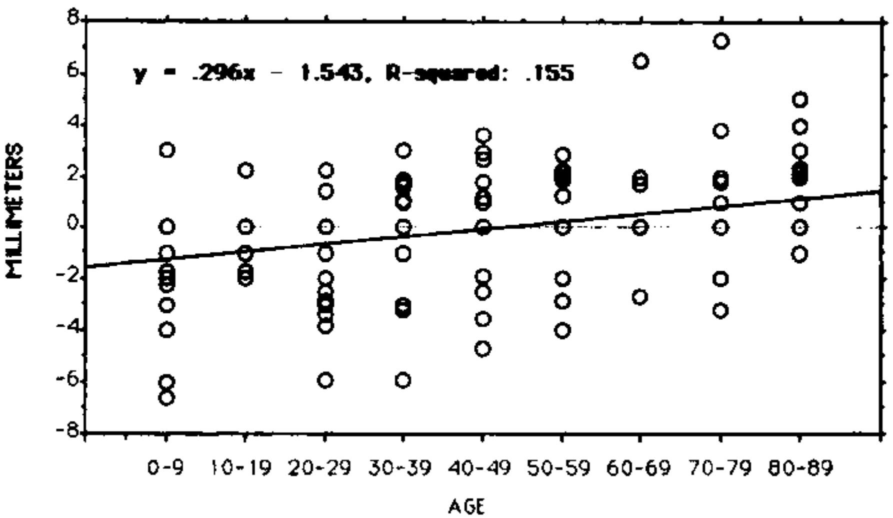
Age range (years)	Male			Female		
	OS(+)	OS(−)	Total	OS(+)	OS(−)	Total
0–9	3	1	4	0	1	1
10–19	5	13	18	3	6	9
20–29	3	19	22	12	17	29
30–39	4	16	20	6	27	33
40–49	6	20	26	10	27	37
50–59	21	39	60	33	38	71
60–69	18	37	55	31	35	66
70–79	27	23	50	21	20	41
80–90	4	4	8	2	3	5
Total	91	172	263	118	174	292

Table 1

Position of the Cerebellar Tonsils Relative to the Foramen Magnum versus Decade of Life in 221 Patients

Decade of Life	No. of Patients	Mean Distance (mm)	SD	95% CL of Mean
1st	24	−1.5	2.2	−2.4, −0.5
2nd	18	−0.4	1.0	−0.9, 0.8
3rd	26	−1.1	1.9	−1.7, −0.2
4th	30	0.0	1.7	−0.6, 0.6
5th	27	0.1	1.7	−0.7, 0.7
6th	23	0.2	1.6	−0.5, 0.8
7th	26	0.2	1.6	−0.5, 0.8
8th	21	0.6	2.1	−0.2, 1.7
9th	26	1.3	1.8	0.6, 2.0

Note.—Negative values indicate distance below the foramen magnum. CL = confidence limits.



Mikulis, Radiology 1992

Table 2

Suggested Criteria for Ectopia of the Cerebellar Tonsils as a Function of Age

Decade of Life	Distance below Foramen Magnum (mm)
1st	6
2nd to 3rd	5
4th to 8th	4
9th	3

Note.—The indicated distances below the foramen magnum are more than 2 SDs outside the normal range.

Mikulis, Radiology 1992

Chiari I

- Most (85%-90% sporadic)

Disorder or Syndrome	Associated clinical anomalies	Chromosome location (gene)	OMIM	Inheritance model
Achondroplasia	Skeletal	4p16.3 (*FGFR3*)	#100800	AD & sporadic
Acromegaly	Endocrine	20q13.32 (*GNAS*)	#102200	unknown
Blepharophimosis	Ophthalmic	3q23 (*FOXL2*)	#110100	AD
Bone marrow failure syndrome 1	Hematologic	4q12 (*SRP72*), 9q22 (*ERCC6L2*)	#614675 #615715	AD & AR
Cleidocranial dysplasia	Skeletal	6p21.1 (*RUNX2*)	#119600	AD
Craniometaphyseal dysplasia	Skeletal	5p15.2 (*ANKH*), 6q22.31 (*GJA1*)	#123000, #218400	AD & AR
Cystic fibrosis	Exocrine	7q31.2 (*CFTR*)	#219700	AR
Epilepsy	Neurologic	8q24	%600131	AD
Fanconi anemia	Hematologic	16q24.3 (*FANCA*)	#227650	AR
Furhmann syndrome	Skeletal, dermatologic & endocrine	3p25.1 (*WNT7A*)	#228930	AR
Goldenhar syndrome	Skeletal	14q32	%164210	AD
Growth hormone deficiency	Endocrine	17q23.3 (*GH1*) Xq22.1 (*BTK*)	#262400 #173100 #307200	AD, AR & X-linked
Hajdu–Cheney syndrome	Skeletal	1p12 (*NOTCH2*)	#102500	AD
Hyper IgE syndrome	Immune	17q21.2 (*STAT3*)	#147060	AD
Hypophosphatemic rickets	Skeletal	Xp22.2-p22.1 (*PHEX*)	#307800	X-linked
Kabuki syndrome	Miscellaneous (FF, ID, skeletal, etc.)	12q13.12 (*KMT2D*)	#147920	AD
Klippel-Feil syndrome	Skeletal	8q22.1 (*GDF6*), 17q21.31 (*MEOX1*), 12p13.31 (*GDF3*)	#118100 #214300 #613702	AD, AR & sporadic
Leopard syndrome	Miscellaneous (dermatologic, etc.)	12q24.13 (*PTPN11*)	#151100	AD
Neurofibromatosis type I	Dermatologic	17q11.2 (*NF1*)	#162200	AD
Noonan syndrome	Miscellaneous (skeletal, FF, etc.)	12q24.13 (*PTPN11*)	#163950	AD

Chiari I

- Most (85%-90% sporadic)

Osteopetrosis	Skeletal	11q13.2 (*LRP5*), 13q14.11 (*TNFS11*), 16p13.3 (*CLCN7*)	#166600, #611490 #607634	AD, AR & X-linked	72-74
Paget disease of bone	Skeletal	18q22.1 (*TNFRSF11A*), 5q35 (*SQSTM1*), 5q31 (*PDB4*) 8q24.12 (*TNFRSF11B*) 1q21.3 (*ZNF687*)	#602080 #167250 %606263 #239000 #616833	AD & AR	75, 76
Pseudohypoparathyroidism type 1A	Endocrine	20q13.32 (*GNAS*)	#103580	AD	77
Renal-coloboma syndrome	Ophthalmic & renal	10q24.3-q25.1 (*PAX-2*)	#120330	AD	78
Rubenstein-Taybi syndrome	Miscellaneous (ID, skeletal, FF, etc.)	16p13.3 (*CREBBP*), 22q13.2 (*EP300*)	#180849, #613684	AD & sporadic	53, 79, 80
Scoliosis	Skeletal	19p13.3, 17p11.2 8q12 9q31.2-q34.2	%181800 %607354 %608765 %612238	AD	81-85
Severe combined pituitary hor-mone deficiency	Endocrine	3p11.2 (*POU1F1*), 5q35.3 (*PROP1*), 9q34.3 (*LHX3*), 1q25.2 (*LHX4*)	#613038 #262600 #221750 #262700	AD & AR	53
Spondylo-epiphyseal dysplasia tarda	Skeletal	unknown	#271600	AD, AR & X-linked	86
Townes-Brocks syndrome	Skeletal & gastrointestinal	16q12.1 (*SALL1*)	#107480	AD	87
Velocardiofacial syndrome	Miscellaneous (skeletal, neurologic, etc.)	22q11(*TBX1*)	#192430	AD	88
William's syndrome	Miscellaneous (ID, FF, cardiovascular, etc.)	7q11.2 (Deletion of aprox 28 genes including: *LIMK1*, *RFC2*, *WBSCR1*, *WBSCR2*)	#194050	AD	89, 90

Causes of "Tonsillar Ectopia"

Cranial constriction	Spinal cord tethering	Cranial settling	Intracranial hypertension	Intraspinal hypotension
Chiari malformation type I	Tethered cord syndrome	HDCT with OAAJI	Hydrocephalus	Prolonged LPS
Craniosynostosis	Chiari malformation type II	Posttraumatic OAAJI	PCF cysts, tumors	CSF leaks
Achondroplasia		Osteogenesis imperfecta	Subdural hematoma	Dural ectasia
Acromegaly			Intracranial mass lesions	
Paget's disease				

Abbreviations: *OAAJI* occipitoatlantoaxial joint instability, *HDCT* hereditary disorders of connective tissue, *LPS* lumboperitoneal shunt, *PCF* posterior cranial fossa, *CSF* cerebrospinal fluid

	Occipital bone size	PCFV	FM
CM-I (classical type)	Small	Small	Small
CM-I/craniosynostosis	Small	Small	Small
CM-II/(with myelodysplasia)	Small	Small	Large
CM-I/tethered cord syndrome	Normal	Normal	Large
CM-I/cranial settling	Normal	Normal	Normal
CM-I/intracranial space-occupying lesions	Normal	Normal	Normal
CM-I/lumboperitoneal shunt	Normal	Normal	Normal

Abbreviations: *PCFV* posterior cranial fossa volume, *FM* foramen magnum, *CM-I* Chiari malformation type I, *CM-II* Chiari malformation type II

Milhorat et al., 2010

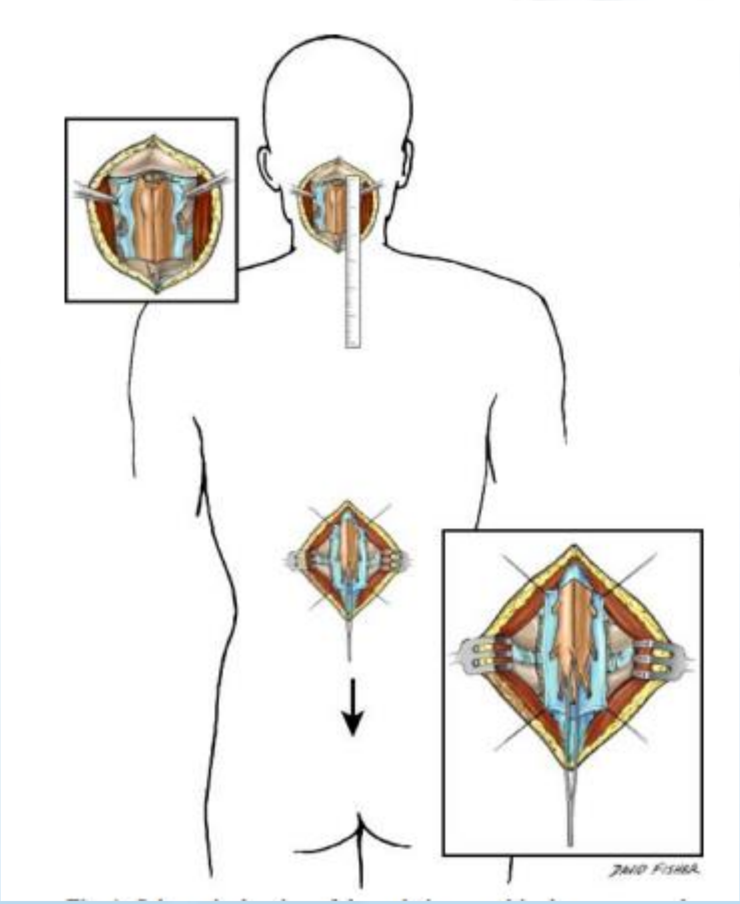
Pathophysiology of CM

1. Caudal Traction

2. Paraaxial mesoderm insufficiency
   - Marin-Parilla : Vitamin A induced
   - Goel : Small posterior fossa
   - Nishikawa : underdeveloped occipital bone

3. Theory of Instability
   - CM a manifestation of atlantoaxial instability.

Caudal Traction



12 fresh adult cadavers <6h postmortem

Distal tension to conus results in negligible movement (<1mm) of caudal brain stem and cervical spinal cord.

No movement of tonsils.

No indication to section filum for CM1

(not so self-evident)

Tubb et al.

Presentation

Table 4. Incidence of Symptoms in 3 Key Articles

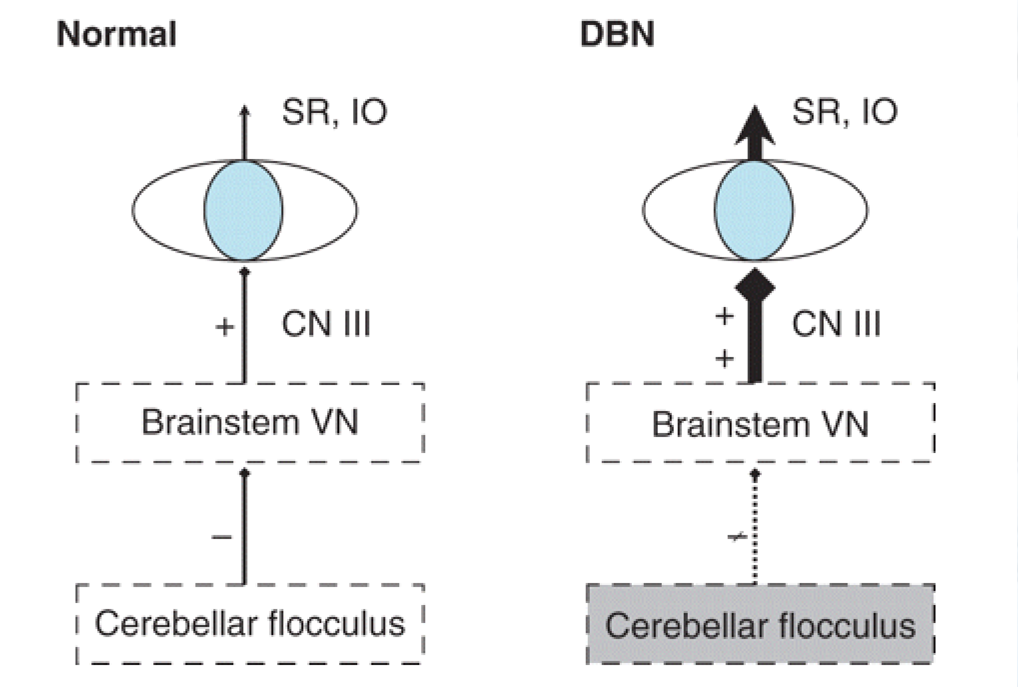
Symptom	Killeen et al.[5] (n = 47)	Chavez et al.[4] (n = 177)	Hayhurst et al.[18] (n = 96)	Combined
Cough headache only	9 (19%)	67		76/224 (34%)
Migraine headache only	14 (30%)	34		48/224 (21%)
Both cough and migraine headache	18 (38%)	31		49/224 (22%)
Headache of any type	41 (87%)	132	63 (66%)	236/320 (74%)
Paresthesia	31 (70%)	66	14 (15%)	111/320 (35%)
Cerebellar symptoms	26 (55%)	36	13 (14%)	75/320 (23%)
Dysphagia	10 (21%)			10/47 (21%)
Dysphagia or apnea		16		16/177 (9%)
Nausea	7 (15%)	23		30/224 (13%)
Drop attacks		4	7	11/273 (4%)
Cranial nerve dysfunction			14 (15%)	14/96 (15%)

Langridge et al., 2017

Downbeat Nystagmus

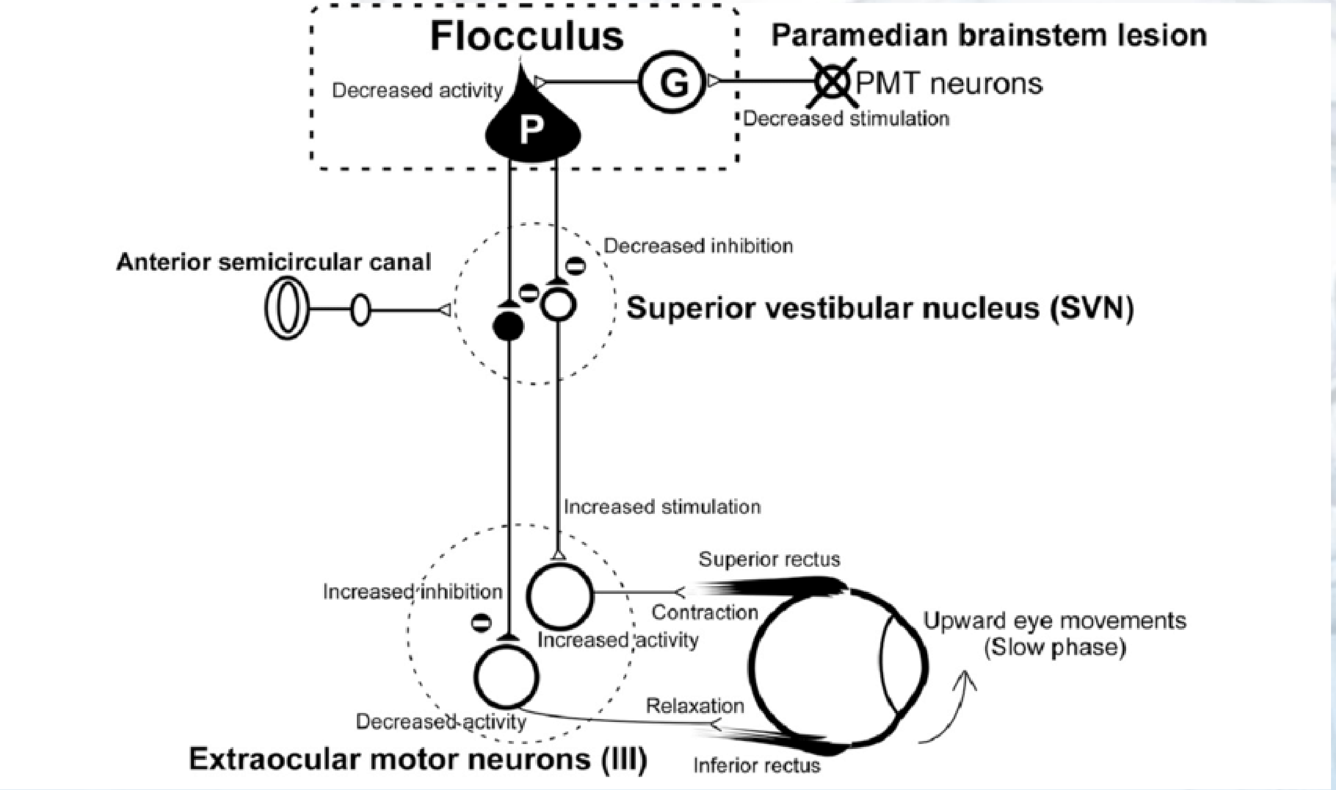
- Abnormal upward ocular drift
- Corrective downward rapid saccade.
- DDX:
  - Chiari malformations,
  - Cerebellar degeneration
  - Drug intoxication,
  - Episodic ataxia type 2

Downbeat Nystagmus



- Dysinhibition of superior vestibular nucleus causes hyperactivity of elevator muscles motor neurons.
- Downward system is unchanged resulting in upward slow eye deviation with corrective downward saccades.

Downbeat Nystagmus



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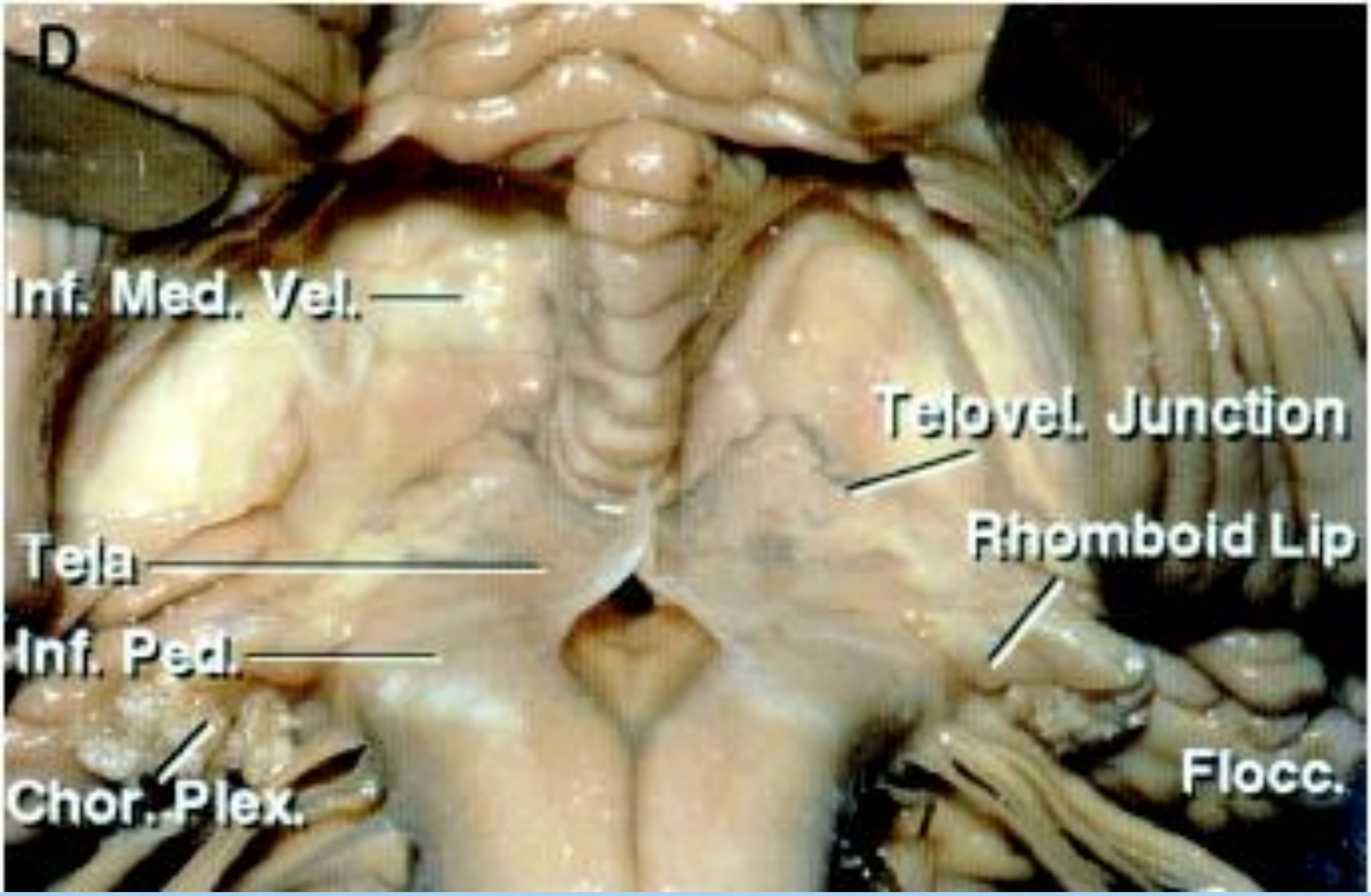

Rhoton, Neurosurgery 2000

Common Incidental Finding

	Number with abnormality	Number needed to scan	Prevalence (%) (95% CI)
Neoplasia (n=19 559)			
Meningioma	72	345	0.29 (0.13 to 0.51)
Pituitary adenoma	27	667	0.15 (0.09 to 0.22)
Low grade glioma	8	2000	0.05 (0.02 to 0.09)
Acoustic neuroma	5	3333	0.03 (0.01 to 0.06)
Lipoma	6	2500	0.04 (0.02 to 0.07)
Epidermoid	3	3333	0.03 (0.01 to 0.06)
Unspecified neoplasm	14	1111	0.09 (0.03 to 0.17)
Any neoplastic incidental finding (n=19 559)	135	143	0.70 (0.47 to 0.98)
Structural vascular abnormalities (n=15 559)			
Aneurysm	67	286	0.35 (0.13 to 0.67)
Cavernous malformation	23	625	0.16 (0.10 to 0.23)
Arteriovenous malformation	7	2000	0.05 (0.01 to 0.10)
Inflammatory lesions (n=15 559)			
Definite demyelination	9	1667	0.06 (0.02 to 0.15)
Possible demyelination	4	3333	0.03 (0.00 to 0.07)
Cysts (n=15 559)			
Arachnoid cyst	99	200	0.50 (0.21 to 0.87)
Colloid cyst	2	2500	0.04 (0.01 to 0.07)
Other abnormalities (n=15 559)			
Chiari I malformation	71	417	0.24 (0.04 to 0.58)
Hydrocephalus	15	1000	0.10 (0.03 to 0.19)
Extra-axial collection	4	2500	0.04 (0.01 to 0.07)
Any non-neoplastic incidental finding* (n=15 559)	375	50	2.00 (1.13 to 3.10)

Prevalence of some incidental findings (*excluding white matter hyperintensities, microbleeds, and silent infarcts) on brain magnetic resonance imaging

(Morris & al., 2009, BMJ)

Common Incidental Finding

5 Don't do routine surveillance imaging for incidentally discovered Chiari I malformation without syringomyelia.

Chiari I malformation, defined as cerebellar tonsillar herniation greater than or equal to 5mm below the foramen magnum on MRI brain, is a frequent incidental finding in children, with an estimated prevalence of 1 to 3%. The vast majority of children with incidentally discovered, asymptomatic Chiari I malformations without syringomyelia have no clinically significant progression of tonsillar descent on routine follow-up, and symptom development is often unassociated with radiographic change. Radiographic follow-up in the absence of new symptomatology is therefore unnecessary.

Choosing Wisely Canada™

Sources:

Gupta SN, et al. Spectrum of intracranial incidental findings on pediatric brain magnetic resonance imaging: What clinician should know? World J Clin Pediatr. 2016 Aug 8;5(3):262-72. PMID: 27610341.

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Chiari and Syrinx

- Described by Russell and Donald in 1935
- Incidence of syrinx ranges from 20 to 75%.
- Syrinx formation is often associated with myelopathic symptoms in a distribution dependent on the spinal cord level involved
  - **____________, ____________ sensory loss**

Chiari and Syrinx

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- Syrinx formation is often associated with myelopathic symptoms in a distribution dependent on the spinal cord level involved
  - **Suspended, dissociated sensory loss**

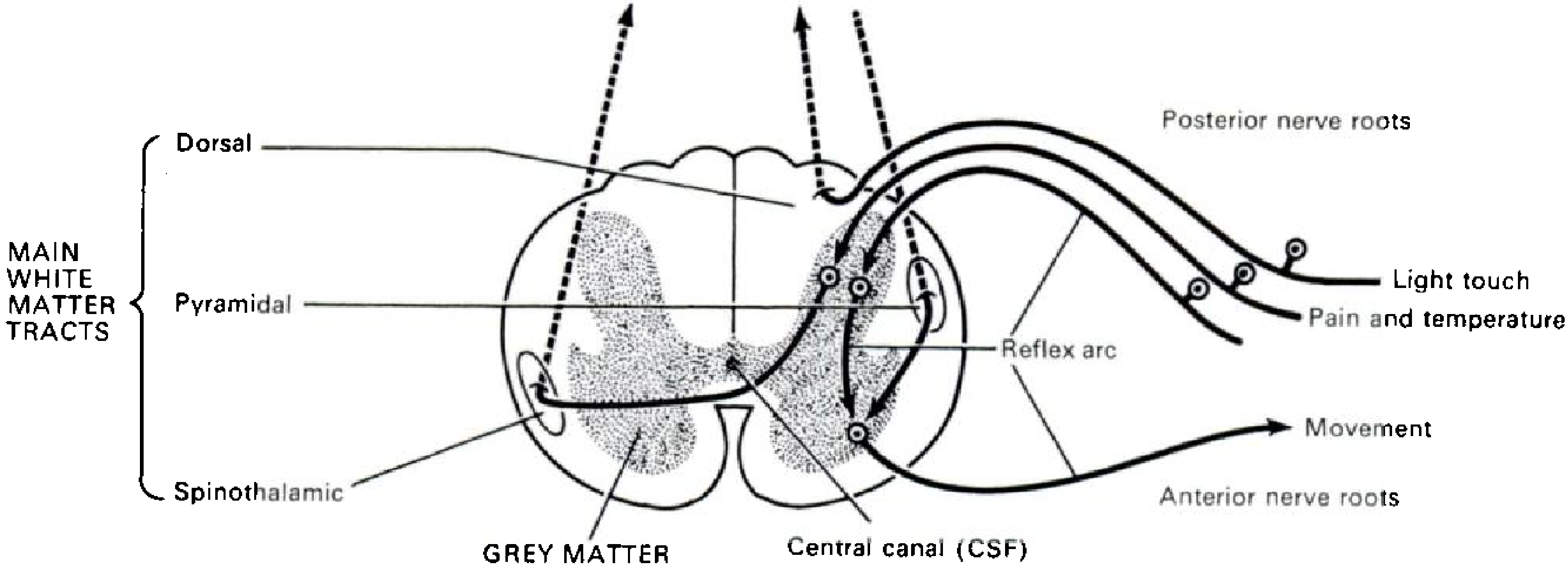


Fig. 1

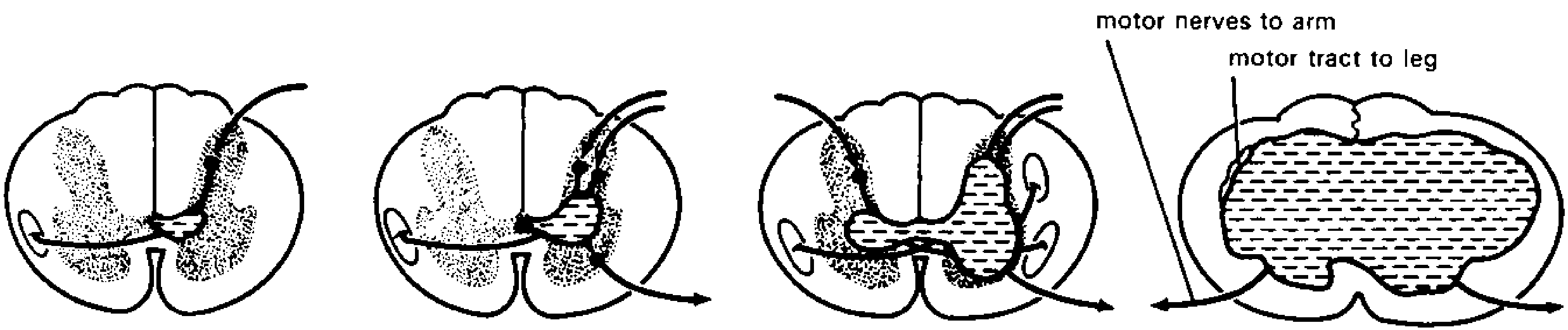
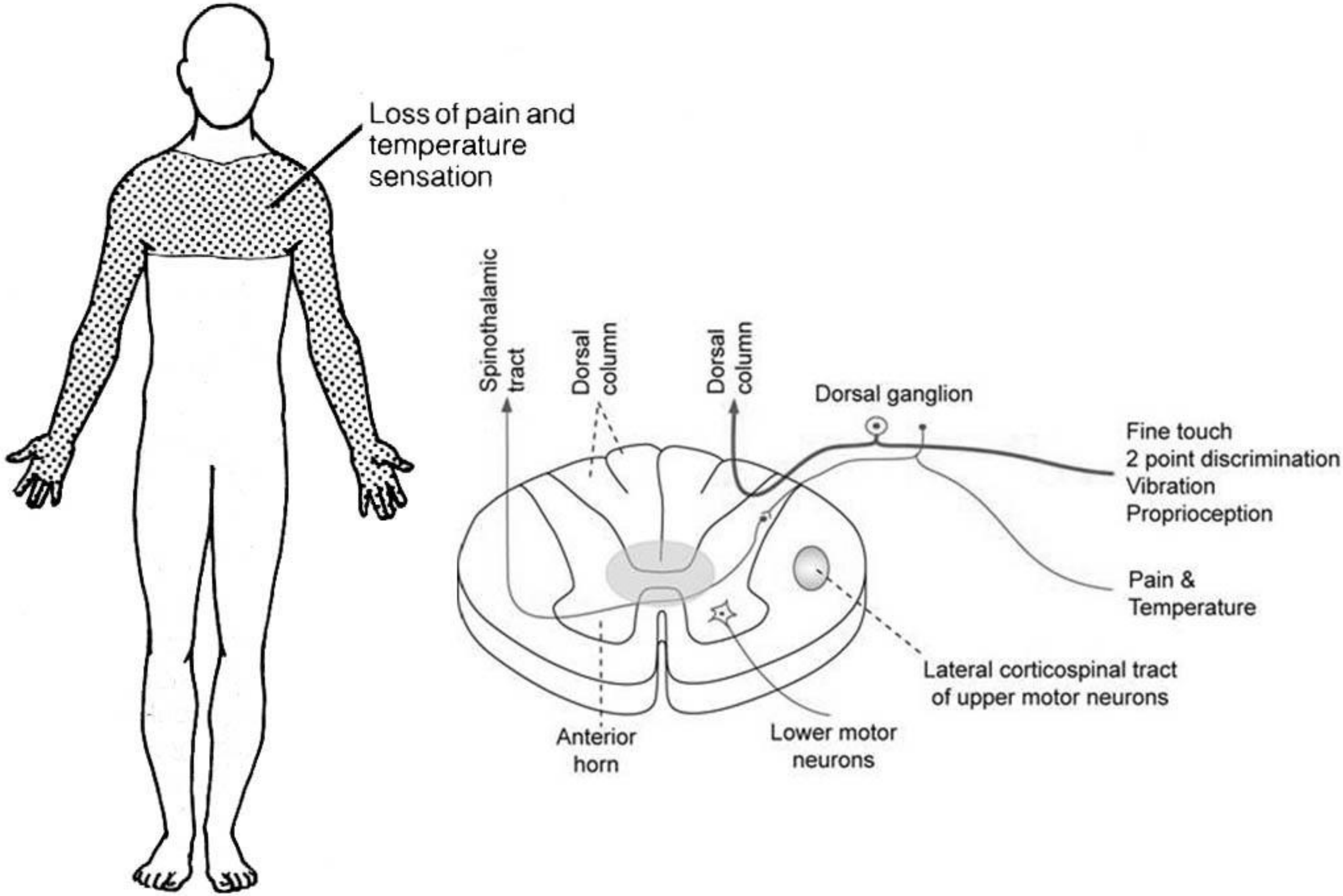


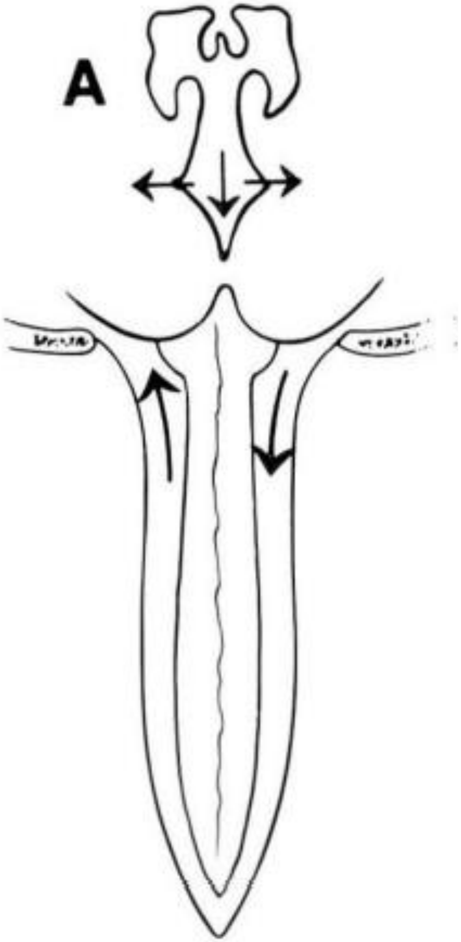
Fig. 2 Fig. 3 Fig. 4 Fig. 5

Relationship between the progression of syringomyelia cavities and nervous system damage. Figure 1—Functions of the normal spinal cord



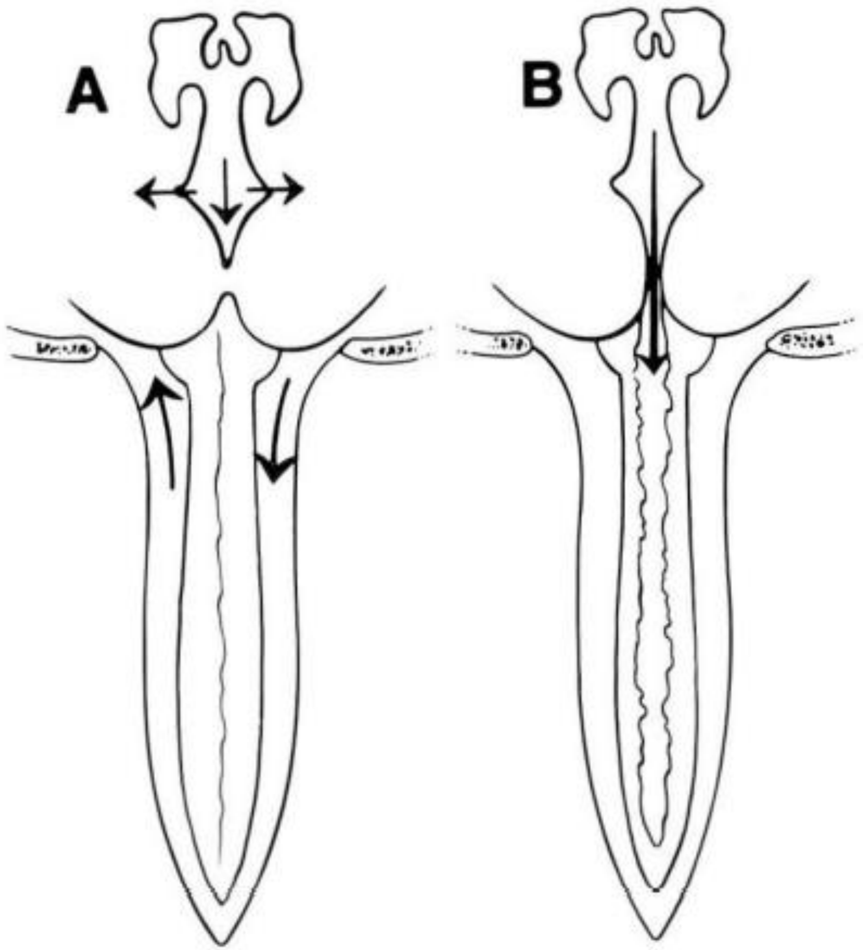
Syringomyelia of the Cervical Cord

Pathophysiology of Syrinx



Pillay et al., 1992

Pathophysiology of Syrinx

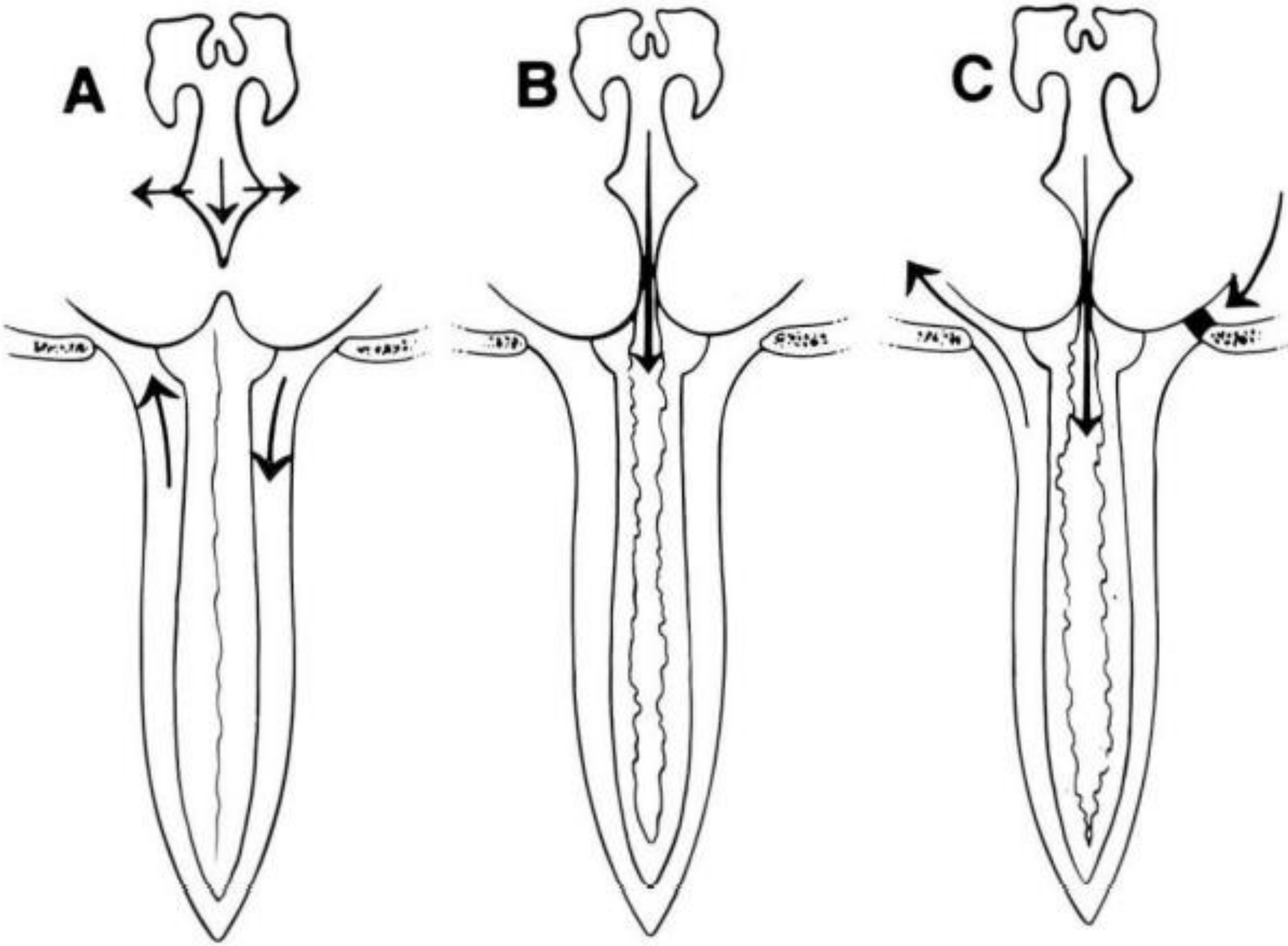


Gardner's Hydrodynamic Theory:

"Bering pulsations";
Obstruction of V4 outlet channels the CSF pulse wave through the obex into central canal of the spine (water hammer effect).

Pillay et al., 1992

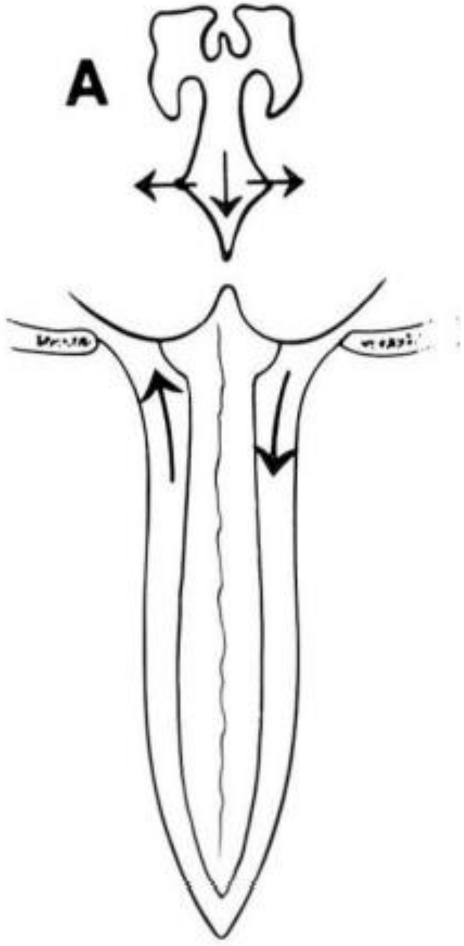
Pathophysiology of Syrinx



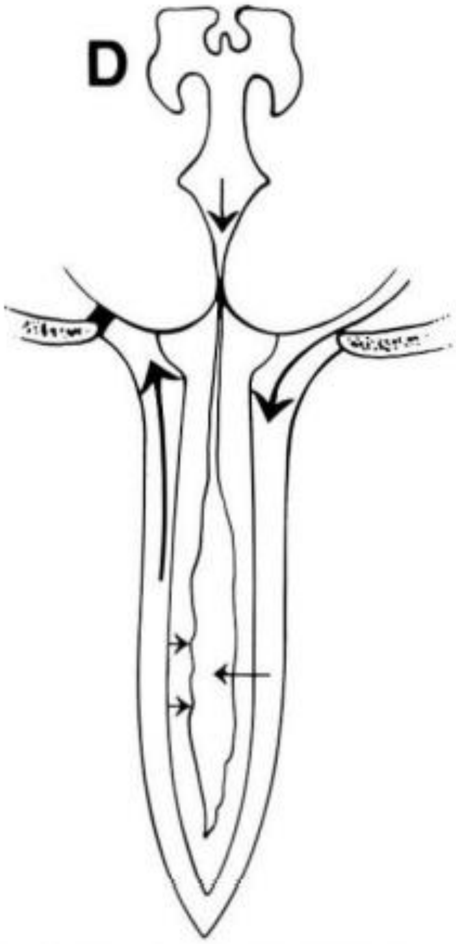
William's Theory:
FM obstruction leads to craniospinal pressure gradient causing hindbrain impaction – Post Valsalva rebound one-way valve "slosh effect"

Pillay et al., 1992

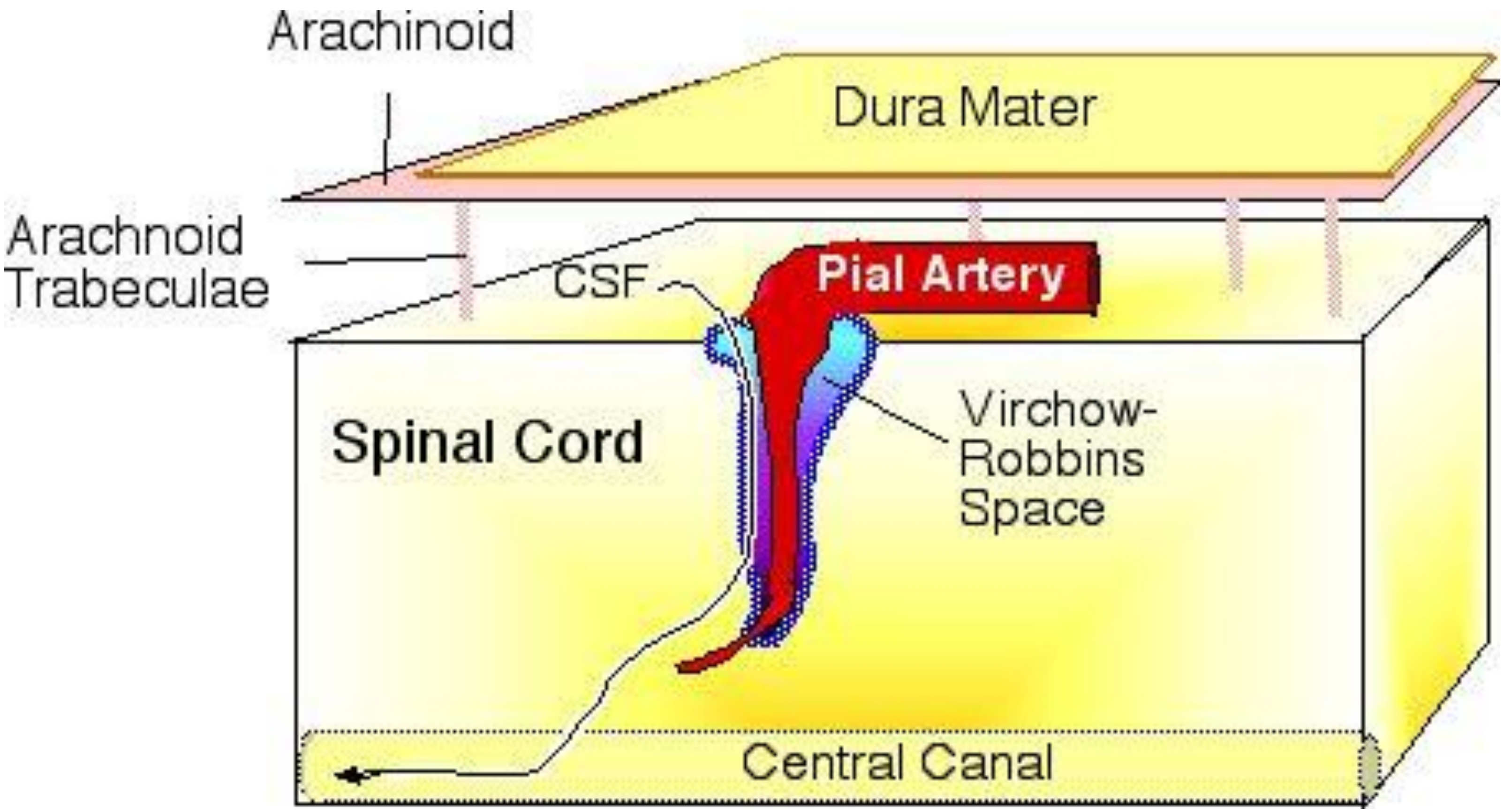
Pathophysiology of Syrinx



Ball, Dayan, Aboulker:
FM pathology leads to increased intraspinal pressure forcing fluid into SC through Virchow-Robin spaces or nerve roots.

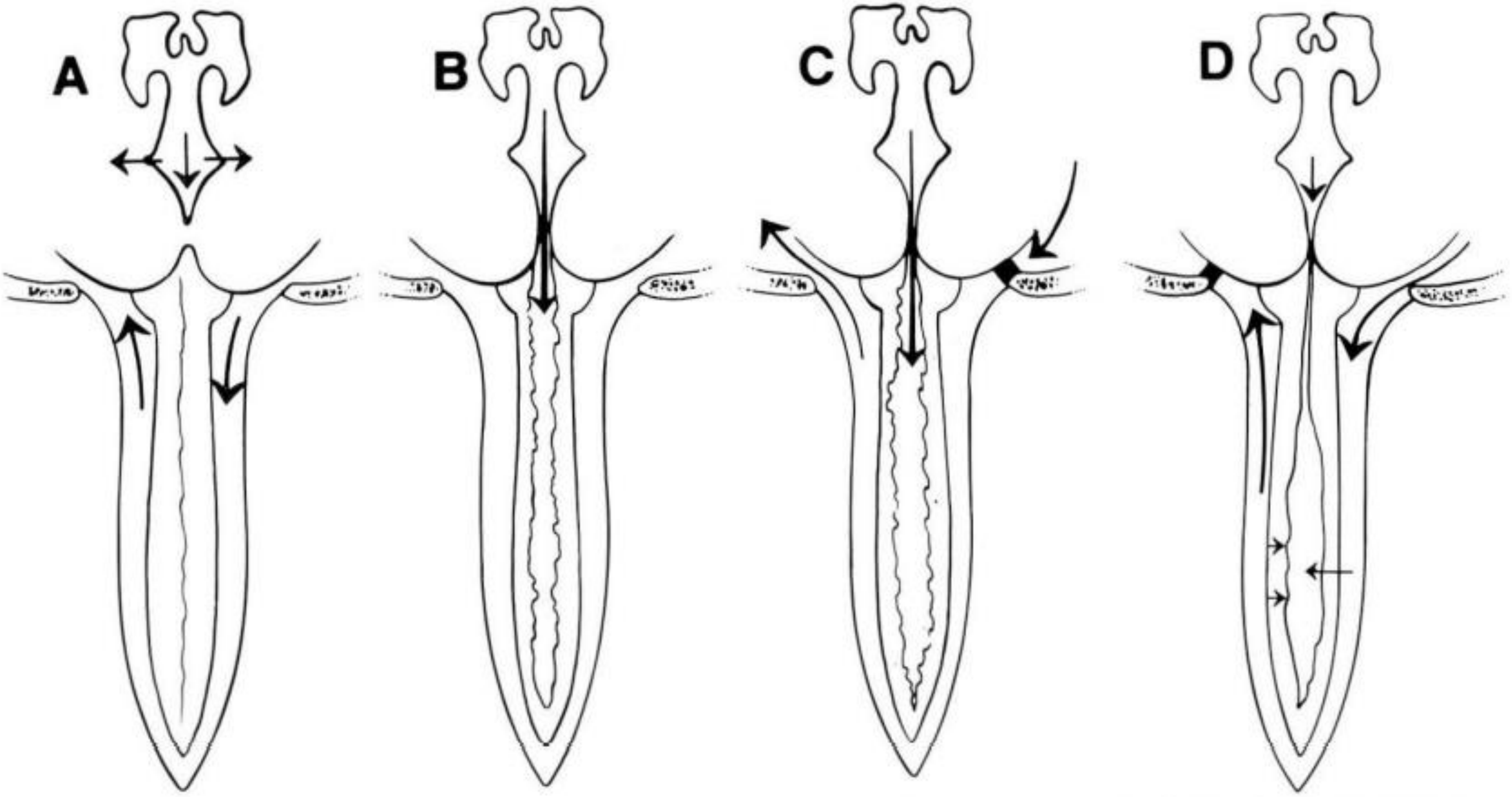


Pillay et al., 1992



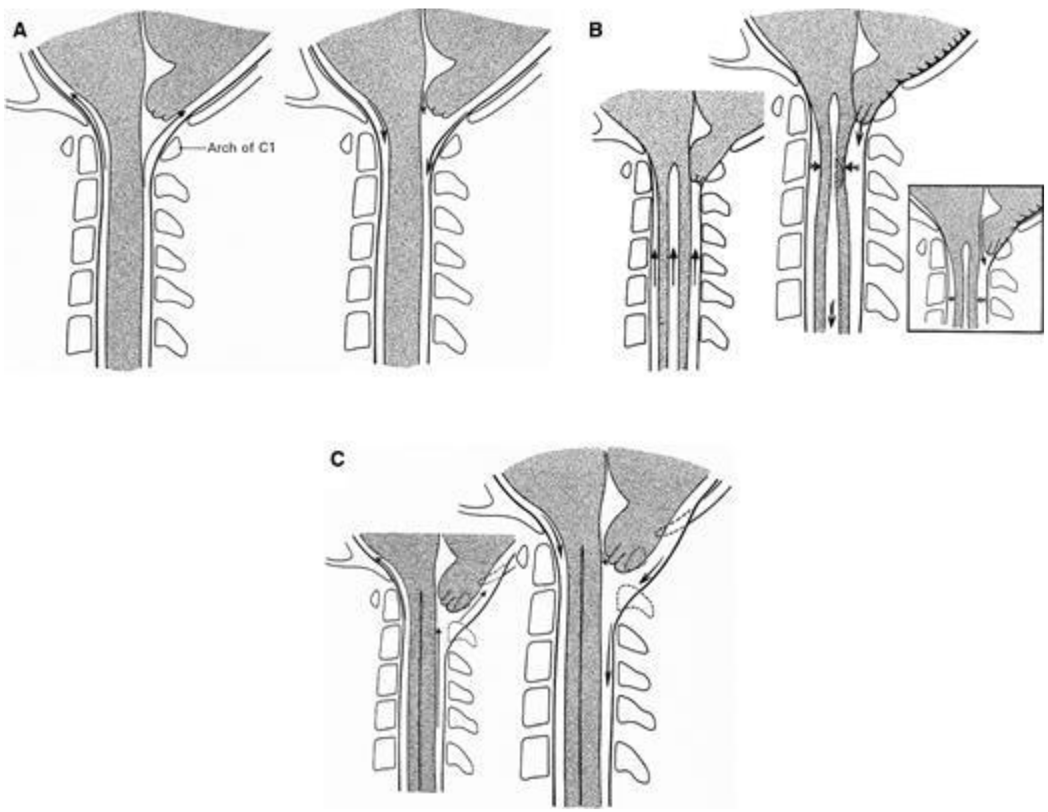
Arachinoid
Dura Mater
Arachnoid
Trabeculae
CSF
Pial Artery
Virchow-
Robbins
Space
Spinal Cord
Central Canal

Pathophysiology of Syrinx



Pillay et al., 1992

Syrinx Expansion: Oldfield



Oldfield:

Normal:

Systole – CSF to SAS

Diastole – CSF flows rostral

With obstruction:

Systole – tonsillar descent

Increases SAS pressure propelling fluid inferiorly and expanding syrinx.

Chiari and Scoliosis

JNS PEDIATRICS

CLINICAL ARTICLE

J Neurosurg Pediatr 24:520–527, 2019

Radiological and clinical predictors of scoliosis in patients with Chiari malformation type I and spinal cord syrinx from the Park-Reeves Syringomyelia Research Consortium

Jennifer M. Strahle, MD,[1] Rukayat Taiwo, MD,[1] Christine Averill, MD,[1] James Torner, MS, PhD,[2] Chevis N. Shannon, DrPH, MBA, MPH,[3] Christopher M. Bonfield, MD,[3] Gerald F. Tuite, MD,[4] Tammy Bethel-Anderson,[1] Jerrel Rutlin,[5] Douglas L. Brockmeyer, MD,[6] John C. Wellons III, MD, MSPH,[3] Jeffrey R. Leonard, MD,[7] Francesco T. Mangano, DO,[8] James M. Johnston, MD,[9] Manish N. Shah, MD,[10] Bermans J. Iskandar, MD,[11] Elizabeth C. Tyler-Kabara, MD, PhD,[12] David J. Daniels, MD, PhD,[13] Eric M. Jackson, MD,[14] Gerald A. Grant, MD,[15] Daniel E. Couture, MD,[16] P. David Adelson, MD,[17] Tord D. Alden, MD,[18] Philipp R. Aldana, MD,[19] Richard C. E. Anderson, MD,[20] Nathan R. Selden, MD, PhD,[21] Lissa C. Baird, MD,[21] Karin Bierbrauer, MD,[8] Joshua J. Chern, MD, PhD,[22] William E. Whitehead, MD,[23] Richard G. Ellenbogen, MD,[24] Herbert E. Fuchs, MD, PhD,[25] Daniel J. Guillaume, MD,[26] Todd C. Hankinson, MD, MBA,[27] Mark R. Iantosca, MD,[28] W. Jerry Oakes, MD,[9] Robert F. Keating, MD,[29] Nickalus R. Khan, MD,[30] Michael S. Muhlbauer, MD,[30] J. Gordon McComb, MD,[31] Arnold H. Menezes, MD,[32] John Ragheb, MD,[33] Jodi L. Smith, PhD, MD,[34] Cormac O. Maher, MD,[35] Stephanie Greene, MD,[12] Michael Kelly, MD,[36] Brent R. O'Neill, MD,[27] Mark D. Krieger, MD,[31] Mandeep Tamber, MD, PhD,[37] Susan R. Durham, MD, MS,[38] Greg Olavarria, MD,[39] Scellig S. D. Stone, MD, PhD,[40] Bruce A. Kaufman, MD,[41] Gregory G. Heuer, MD, PhD,[42] David F. Bauer, MD,[43] Gregory Albert, MD, MPH,[44] Jeffrey P. Greenfield, MD, PhD,[45] Scott D. Wait, MD,[46] Mark D. Van Poppel, MD,[46] Ramin Eskandari, MD,[47] Timothy Mapstone, MD,[48] Joshua S. Shimony, MD, PhD,[5] Ralph G. Dacey Jr., MD,[1] Matthew D. Smyth, MD,[1] Tae Sung Park, MD,[1] and David D. Limbrick Jr., MD, PhD[1]

TABLE 3. Demographic information and radiological variables of patients with or without scoliosis

	Scoliosis (n = 260)	No Scoliosis (n = 49)	p Value	OR (scoliosis)	95% CI	Multivariate OR	95% CI
Age at Chiari diagnosis, yrs	10.36	9.55	0.19	1.05	0.98–1.14		
Female sex	172 (66%)	30 (61%)	0.50	1.24	0.66–2.32		
Tonsil position, mm*	12.49	12.03	0.26	0.98	0.91–1.05		
Syrinx width, mm	8.66	6.25	<0.0001	1.25	1.13–1.38		
Syrinx length, no. of levels	10.6 levels	7.4 levels	<0.0001	1.18	1.1–1.28	1.127	1.036–1.227
Cervical cranial extent of syrinx	244 (93.8%)	39 (79.6%)	0.001	3.91	1.66–9.23		
Holocord syrinx	131 (50.4%)	8 (16.3%)	<0.0001	5.16	2.32–11.45	2.998	1.257–7.167
pB–C2, mm	7	6.83	0.40	1.11	0.87–1.42		
Clivoaxial angle, °	144.6	144.2	0.85	1.00	0.98–1.03		
Frontal-occipital horn ratio	0.301	0.308	0.38	1.06	0.9–1.25		

Values are presented as number of patients (%) unless otherwise indicated.

* Millimeters below foramen magnum.

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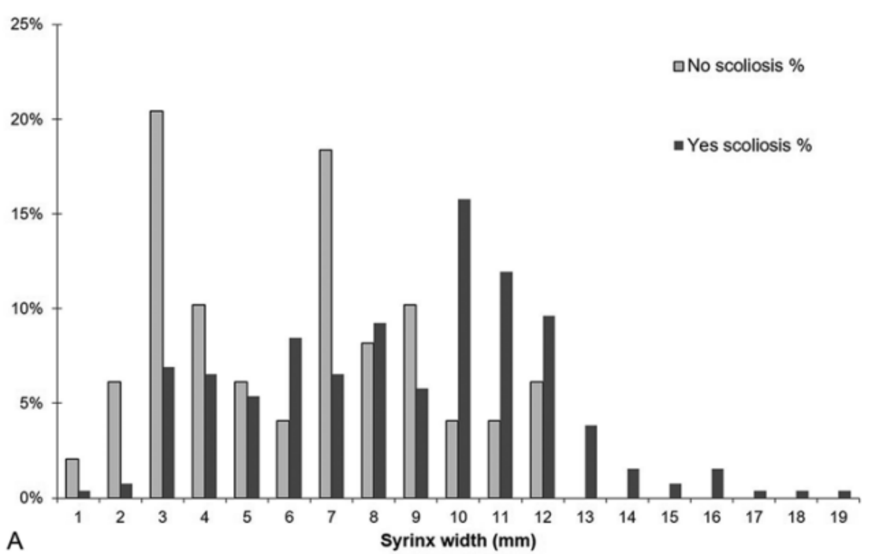

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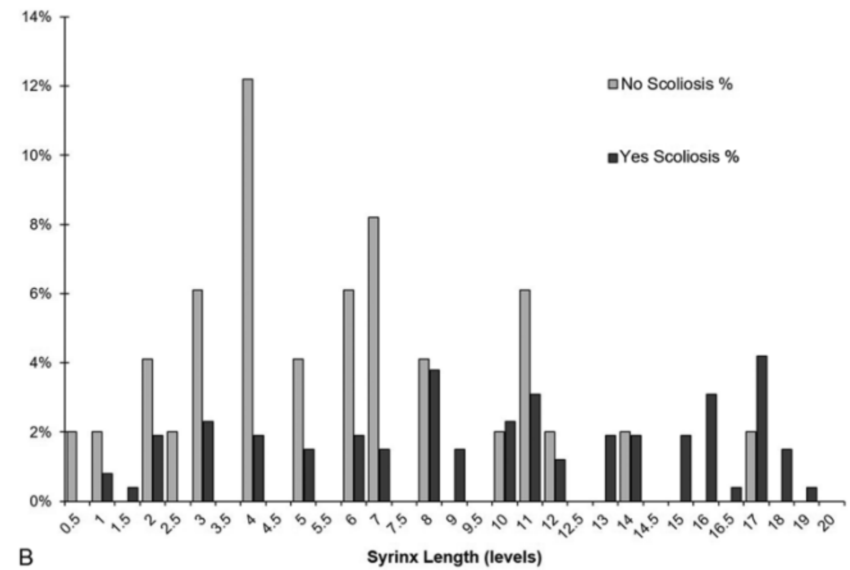

B

Chiari and Scoliosis

- Scoliosis is associated with CM, generally when presenting in combination with syrinx.

- **With Syrinx:** Primary treatment of the Chiari I malformation is the currently accepted rule, often resulting in resolution of the syrinx and cessation of scoliosis curve progression.

- **Without Syrinx:** Insufficient evidence to put forth recommendations for or against the surgical treatment of asymptomatic Chiari I malformations that present with scoliosis in the absence of syrinx.

Surgical Indication

Surgery Should be Considered with :

1. Imaging demonstrating unilateral or bilateral cerebellar tonsil herniation through the foramen magnum;

2. Loss of cerebrospinal fluid (CSF) signal at the craniocervical junction;

3. In the presence of syrinx or attributable symptoms that are refractory to medical management, including recurrent occipital headache, nuchal pain, vertigo, parasthesias, extremity paresis, hyperreflexia, ataxia, or nystagmus

Surgical Treatment

- The primary goal of surgical intervention for CM is to decompress the craniocervical junction, increasing the size of the cisterna magna and restoring normal CSF flow dynamics.
- Options:
  - Suboccipital decompression (SOD) with an appropriate number of cervical laminectomies,
  - Dural opening with patch grafting (duraplasty),
  - Intradural procedures, including lysis of arachnoid adhesions, obex plugging, and cauterization or resection of the cerebellar tonsils.
  - Syringocisternostomy and placement of syringosubarachnoid, syringopleural, syringoperitoneal, thecoperitoneal, and fourth ventricular shunts and stents

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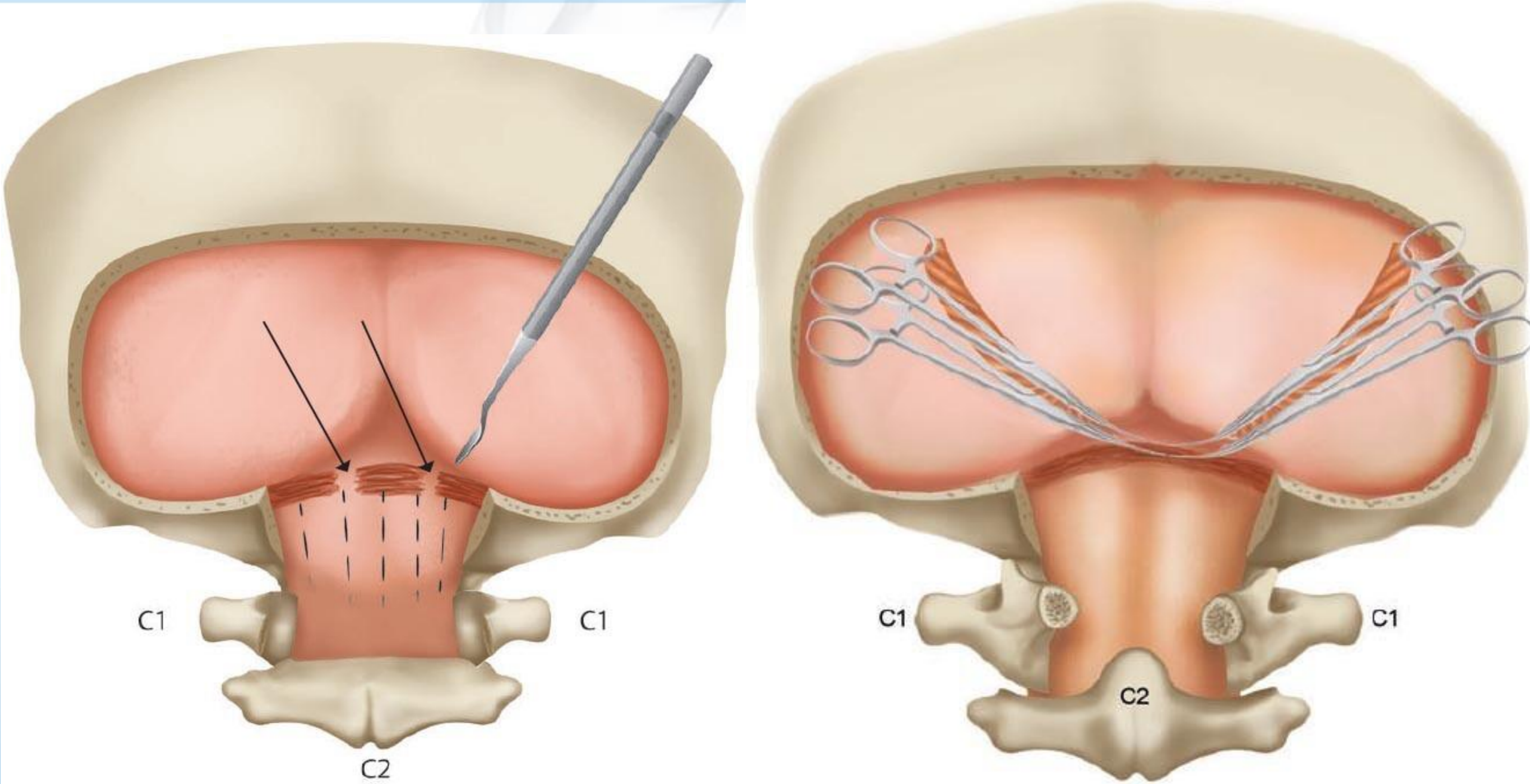
JNS PEDIATRICS
CLINICAL ARTICLE
J Neurosurg Pediatr 28:167–175, 2021

Extradural decompression versus duraplasty in Chiari malformation type I with syrinx: outcomes on scoliosis from the Park-Reeves Syringomyelia Research Consortium

*Brooke Sadler, PhD,[1] Alex Skidmore,[2] Jordan Gewirtz, BS,[2] Richard C. E. Anderson, MD,[17] Gabe Haller, PhD,[2] Laurie L. Ackerman, MD,[4] P. David Adelson, MD,[5] Raheel Ahmed, MD, PhD,[6] Gregory W. Albert, MD,[7] Philipp R. Aldana, MD,[8] Tord D. Alden, MD,[9] Christine Averill, MD,[2] Lissa C. Baird, MD,[10] David F. Bauer, MD, MPH,[11] Tammy Bethel-Anderson,[2] Karin S. Bierbrauer, MD,[12] Christopher M. Bonfield, MD,[43] Douglas L. Brockmeyer, MD,[13] Joshua J. Chern, MD, PhD,[14] Daniel E. Couture, MD,[15] David J. Daniels, MD, PhD,[16] Brian J. Dlouhy, MD,[39] Susan R. Durham, MD,[18] Richard G. Ellenbogen, MD,[19] Ramin Eskandari, MD,[20] Herbert E. Fuchs, MD, PhD,[21] Timothy M. George, MD,[22] Gerald A. Grant, MD,[23] Patrick C. Graupman, MD,[24] Stephanie Greene, MD,[25] Jeffrey P. Greenfield, MD, PhD,[26] Naina L. Gross, MD,[27] Daniel J. Guillaume, MD,[28] Todd C. Hankinson, MD,[29] Gregory G. Heuer, MD, PhD,[30] Mark Iantosca, MD,[31] Bermans J. Iskandar, MD,[6] Eric M. Jackson, MD,[32] Andrew H. Jea, MD,[4] James M. Johnston, MD,[33] Robert F. Keating, MD,[34] Nickalus Khan, MD,[36] Mark D. Krieger, MD,[37] Jeffrey R. Leonard, MD,[38] Cormac O. Maher, MD,[3] Francesco T. Mangano, DO,[12] Timothy B. Mapstone, MD,[27] J. Gordon McComb, MD,[37] Sean D. McEvoy, MD,[2] Thanda Meehan, RN, BSN,[2] Arnold H. Menezes, MD,[39] Michael Muhlbauer, MD,[36] W. Jerry Oakes, MD,[33] Greg Olavarria, MD,[40] Brent R. O'Neill, MD,[29] John Ragheb, MD,[41] Nathan R. Selden, MD, PhD,[10] Manish N. Shah, MD,[42] Chevis N. Shannon, DrPH,[43,47] Jodi Smith, MD, PhD,[4] Matthew D. Smyth, MD,[2] Scellig S. D. Stone, MD, PhD,[44] Gerald F. Tuite, MD,[45] Scott D. Wait, MD,[46] John C. Wellons III, MD, MSPH,[43,47] William E. Whitehead, MD,[11] Tae Sung Park, MD,[2] David D. Limbrick Jr., MD, PhD,[1,2] and Jennifer M. Strahle, MD[1,2,35]



Atlantooccipital membrane – most compression

TABLE 2. Syrinx characteristics for extradural decompression only and duraplasty

	Extradural Decompression	Duraplasty	p Value
Syrinx width, mm			
Before PFD	7.6	8.9	**0.02**
After PFD	2.2	1.9	0.53
Change	–5.4	–7.0	**0.03**
Syrinx length, vertebral levels			
Before PFD	9.4	10.7	**0.04**
After PFD	9.6	8.2	0.15
Change	–0.6	–2.3	0.08
Holocord syrinx, %	34	50	0.10

Boldface type indicates statistical significance.

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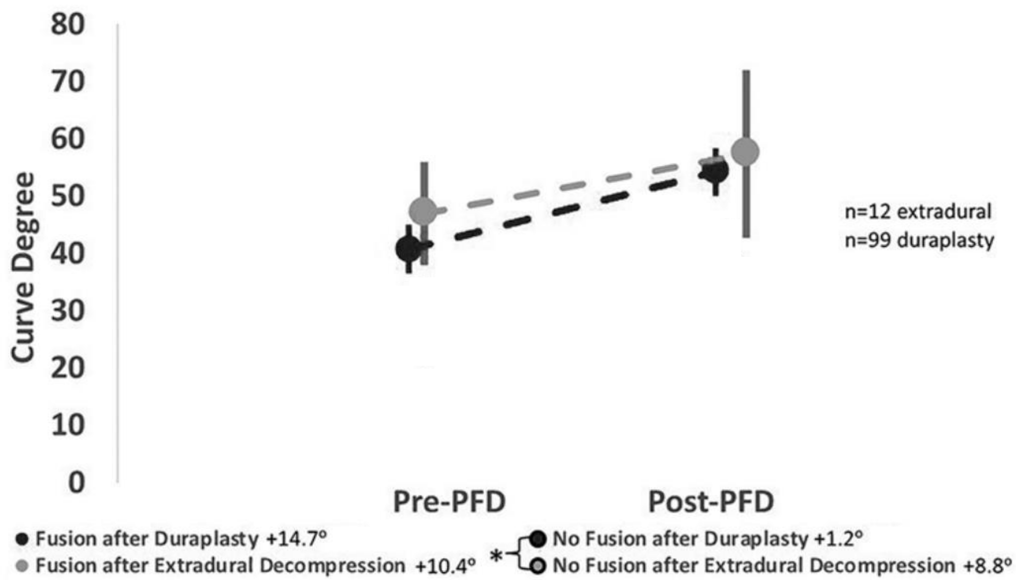
Patients who went on to have a fusion for scoliosis had progression regardless of surgery type



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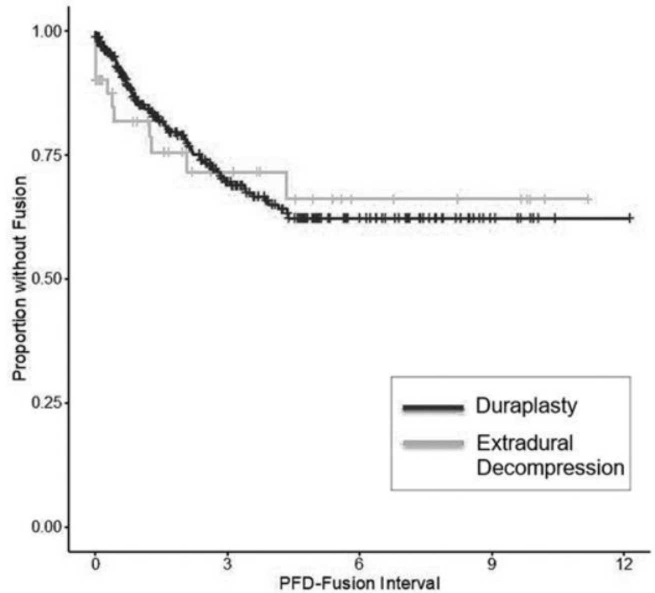
No difference in fusion proportion between two procedures



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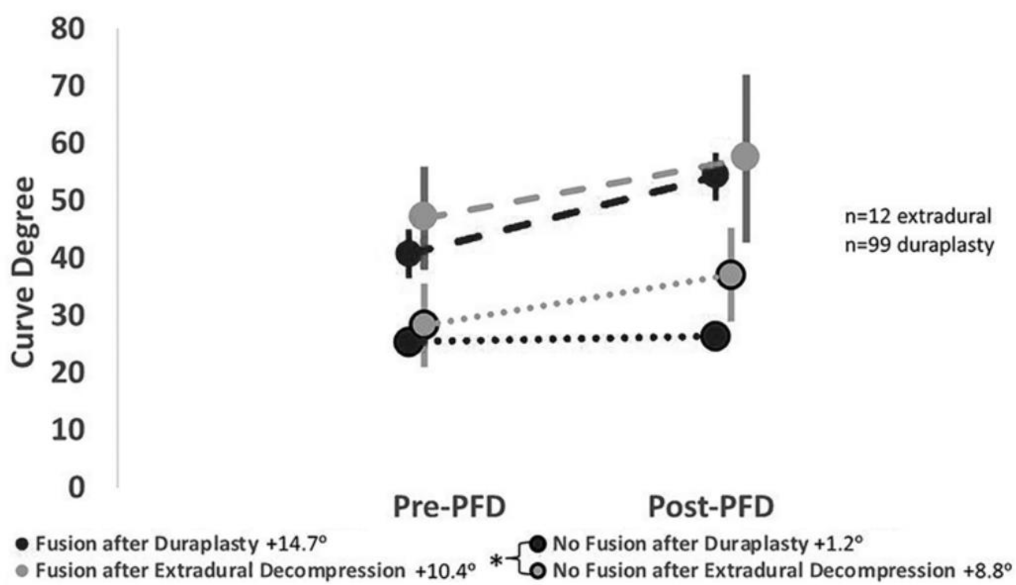
Patients with duraplasty were less likely to have curve progression



Factors associated with syrinx resolution

TABLE 2. Factors associated with syrinx resolution, syrinx regression, or preoperative syrinx diameter by risk-adjusted CPH modeling using follow-up time as event

	HR (95% CI)	p Value
Syrinx resolution (≤2-mm AP diameter)		
PFDD	1.78 (1.00–3.17)	0.049
Younger age at surgery	1.10 (1.04–1.16)	0.0002
Preop syrinx diameter	0.84 (0.68–1.23)	0.081
Female sex	1.76 (1.13–2.75)	0.157
Syrinx resolution (≤3-mm AP diameter)		
PFDD	1.89 (1.22–2.07)	0.014
Younger age at surgery	1.21 (1.11–1.28)	0.001
Female sex	1.01 (0.47–2.46)	0.320
Syrinx resolution (≥50% decrease in AP diameter)		
PFDD	2.04 (1.34–2.69)	0.039
Younger age at surgery	1.08 (1.01–1.22)	0.001
Scoliosis	1.74 (0.89–2.88)	0.100
Syrinx resolution (AP diameter >10 mm)		
PFDD	3.29 (2.88–4.99)	0.009
Younger age at surgery	1.60 (1.29–1.94)	0.001
Syrinx resolution (AP diameter >5 to ≤10 mm)		
PFDD	1.79 (1.08–3.27)	0.041
Younger age at surgery	1.08 (1.05–1.11)	0.020
Female sex	1.99 (0.89–2.88)	0.039
Syrinx resolution (AP diameter ≤5 mm)		
Age at surgery	1.07 (0.98–1.17)	0.457
Scoliosis	1.87 (0.69–5.09)	0.549
Tonsillar descent	0.85 (0.75–0.93)	0.079
Clivus angle	0.99 (0.96–1.02)	0.753
FOHR	0.47 (0.32–0.65)	0.082

- PFDD
- Younger age at surgery
- Initial syrinx size

Hale et al., 2020

Pause and Assess !!

- Contraindications to the primary surgical treatment of children with Chiari I malformation via SOD, with or without duraplasty:

Pause and Assess !!

- Contraindications to the primary surgical treatment of children with Chiari I malformation via SOD, with or without duraplasty:
  - Significant **untreated hydrocephalus**
  - Significant **ventral brainstem compression**
  - Significant **craniocervical instability**

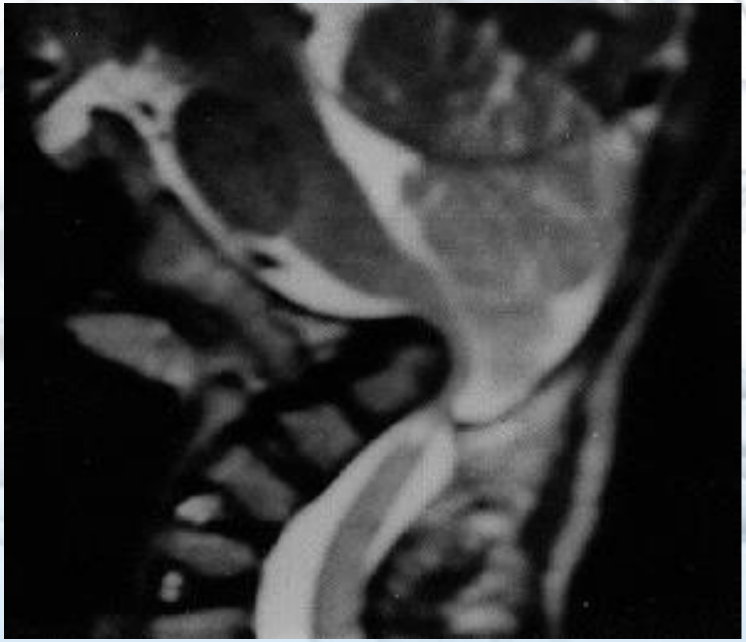
"Complex Chiari" – Associated with skull base anomalies

RESEARCH—HUMAN—CLINICAL STUDIES

Occipital-Cervical Fusion and Ventral Decompression in the Surgical Management of Chiari-1 Malformation and Syringomyelia: Analysis of Data From the Park-Reeves Syringomyelia Research Consortium

Important Associated Anomalies

- Underdeveloped Occiput
- Small posterior fossa
- Short clivus
- Platybasia
- Basilar invagination
- Retroflexed odontoid
- Atlantooccipital assimilation
- Klippel-Feil
- Levo-scoliosis (often with syrinx)



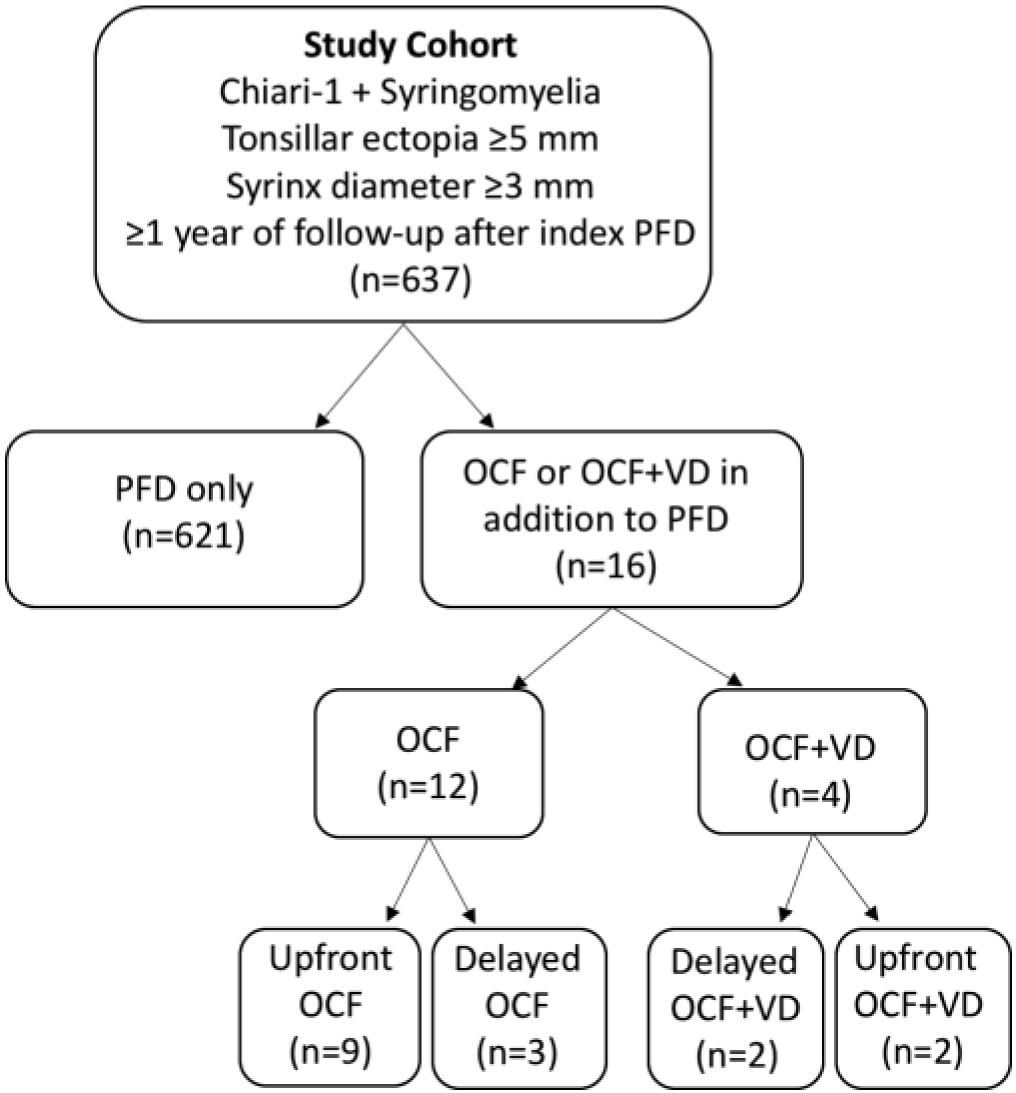


FIGURE. *Flowchart depicting the study cohort.*

TABLE 3. Comparison of Clinical Findings at Initial Treatment[a]

Variable	PFD only[a]	OCF	*P* value	OCF/VD	*P* value
Number in group (n)	621	12		4	
Symptoms					
Headache	346/621 (55.7)	8/12 (66.7)	.564	2/4 (50.0)	1.000
Neck pain	129/621 (20.8)	4/12 (33.3)	.289	0/4 (0.0)	.586
Back pain	101/621 (16.2)	1/12 (8.3)	.701	0/4 (0.0)	1.000
Balance/Gait Ataxia	87/621 (14.0)	6/12 (50.0)	**.004**	3/4 (75.0)	**.010**
Dysphagia	70/621 (11.3)	4/12 (33.3)	**.041**	0/4 (0.0)	1.000
UE weakness	56/621 (9.0)	1/12 (8.3)	1.000	0/4 (0.0)	1.000
LE numbness	55/621 (8.9)	2/12 (16.6)	.295	1/4 (25.0)	.314
LE weakness	45/621 (7.2)	1/12 (8.3)	.599	0/4 (0.0)	1.000
UE pain	42/621 (6.7)	1/12 (8.3)	.573	0/4 (0.0)	1.000
UE numbness	42/621 (6.7)	1/12 (8.3)	.702	0/4 (0.0)	.127
Seizure	36/621 (5.8)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
LE pain	33/621 (5.3)	1/12 (8.3)	.488	0/4 (0.0)	1.000
Apnea	32/621 (5.2)	2/12 (16.6)	.132	1/4 (25.0)	.195
Choking	29/621 (4.7)	1/12 (8.3)	.445	2/4 (50.0)	**.013**
Diplopia	22/621 (3.5)	0/12 (0.0)	1.000	2/4 (50.0)	**.008**
Pulmonary infections	12/621 (1.9)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Tremor	11/621 (1.9)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Hoarseness	10/621 (1.6)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Trunk pain	9/621 (1.4)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Face numbness	9/621 (1.4)	0/12 (0.0)	1.000	1/4 (25.0)	.063
Vocal cord dysfunction	8/621 (1.3)	1/12 (8.3)	.159	0/4 (0.0)	1.000
Face weakness	6/621 (0.9)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Shortness of breath	6/621 (0.9)	0/12 (0.0)	1.000	0/4 (0.0)	–
Prior genetic history	46/621 (7.4)	3/12 (25.0)	.058	0/4 (0.0)	1.000
Neurologic exam					
LE weakness	544/555 (98.0)	11/12 (91.7)	.228	4/4 (100.0)	1.000
UE weakness	543/557 (97.5)	10/12 (83.3)	**.042**	3/4 (75.0)	.103
Pinprick	30/251 (11.9)	1/5 (20.0)	.478	0/1 (0.0)	1.000
Gait instability	61/516 (11.8)	5/8 (62.5)	**.001**	2/4 (0.0)	.074
Hyperreflexia	55/493 (11.2)	5/11 (45.5)	**.005**	2/4 (0.0)	.067
Light touch deficit	35/422 (8.3)	1/8 (12.5)	.506	0/3 (0.0)	1.000
Nystagmus	42/525 (8.0)	3/12 (25.0)	.071	1/4 (25.0)	.288
Clonus	25/315 (7.9)	2/6 (33.3)	.083	0/1 (0.0)	1.000
Weak Gag Reflex	33/436 (7.6)	1/8 (12.5)	.474	2/3 (66.6)	**.018**
Babinski	14/233 (6.0)	1/6 (16.7)	.325	0/1 (0.0)	1.000
Hoffman	12/219 (5.5)	0/3 (0.0)	1.000	1/2 (50.0)	.114
Romberg	8/229 (3.5)	0/4 (0.0)	1.000	0/1 (0.0)	1.000
Dysmetria	9/352 (2.6)	1/5 (20.0)	.133	1/2 (50.0)	.056
Disconjugate gaze	12/620 (1.9)	1/12 (8.3)	.222	1/4 (25.0)	.081
Weak neck rotation	9/464 (1.9)	4/10 (40.0)	**<.001**	0/2 (0.0)	1.000
Facial sensation	7/473 (1.5)	1/11 (0.1)	.169	0/3 (0.0)	1.000
Proprioception	3/253 (1.2)	0/5 (0.0)	1.000	0/4 (0.0)	–
Facial weakness	6/549 (1.1)	0/12 (0.0)	1.000	0/3 (0.0)	1.000
Hoarseness	5/461 (1.1)	0/10 (0.0)	1.000	0/3 (0.0)	1.000
Extraocular palsy	5/620 (0.8)	0/12 (0.0)	1.000	0/4 (0.0)	1.000
Tongue deviation	3/548 (0.5)	1/12 (8.3)	.083	0/2 (0.0)	1.000
eWeak shrug	1/442 (0.2)	0/10 (0.0)	1.000	0/2 (0.0)	1.000

Bold value indicate statistically significant *P* values ($p < .05$)

TABLE 4. Initial Radiologic Findings[a]

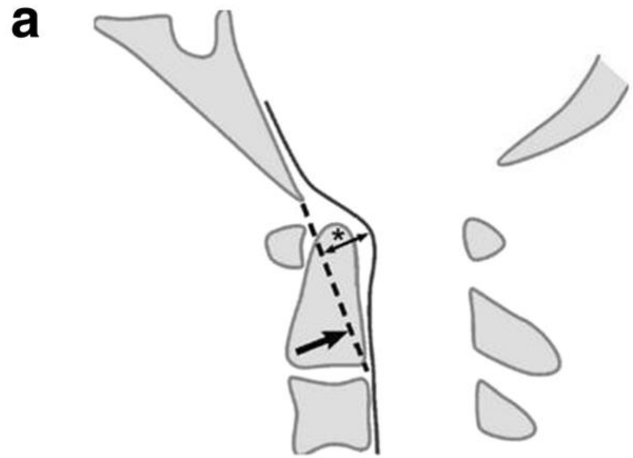
Variable	PFD Only	OCF	OCF Mean Difference	*P* value[e]	OCF/VD	OCF/VD Mean Difference	*P* value[e]	OCF vs OCF/VD *P* value[f]
Number in group (n)	621	12			4			
Age of surgery[b]	10.09 [7.12-16.74]	11.51 [7.10-16.74]	1.42 ± 1.50	.621	12.59 [7.34-15.79]	2.50 ± 2.54	.634	.930
Radiographic findings[c]								
Tonsillar Ectopia (mm)	12.5 ± 4.9	17.3 ± 5.4	−4.8 ± 1.6	**.026**	17.5 ± 5.8	−5.00 ± 2.9	.335	.999
AP Syrinx diameter (mm)	7.5 ± 3.1	7.1 ± 3.0	0.6 ± 0.9	.811	9.5 ± 3.0	−2.1 ± 1.5	.442	.349
Syrinx Levels	8.9 ± 4.6	7.6 ± 5.2	1.2 ± 1.7	.762	10.7 ± 6.1	−1.89 ± 3.5	.864	.735
pB-C2 (mm)	6.9 ± 1.2	8.4 ± 2.2	−1.3 ± 0.7	.164	6.8 ± 0.5	0.1 ± 0.3	.895	.146
pB-C2 ≥ 9 mm[d]	47/510 (9.2)	3/11 (27.3)	–	.079	0/4 (0.0)	–	1.000	.516
Clivoaxial angle (°)	145.5 ± 12.4	128.8 ± 15.3	16.8 ± 4.4	**.008**	115.0 ± 11.6	30.8 ± 5.8	**.025**	.128
Clivoaxial angle ≤ 125°[d]	31/511 (6.0)	7/11 (63.6)	–	**<.001**	3/4 (75.0)	–	**.001**	1.000
FOR	0.3 ± 0.04	0.3 ± 0.02	0.009 ± 0.007	.501	0.3 ± 0.01	0.02 ± 0.005	.226	.668
Spine Pathology[d]								
Platybasia	12/281 (4.3)	3/10 (30.0)	–	**.011**	0/3 (0.0)	–	1.000	.528
Scoliosis	230/637 (36.1)	4/10 (40.0)	–	1.000	1/3 (33.3)	–	.127	1.000
Klippel Feil	4/283 (1.4)	2/10 (20.0)	–	**.015**	0/3 (0.0)	–	1.000	1.000
Basilar invagination	11/281 (3.9)	6/10 (60.0)	–	**<.001**	3/3 (100.0)	–	**<.001**	.497

Risk Factors:

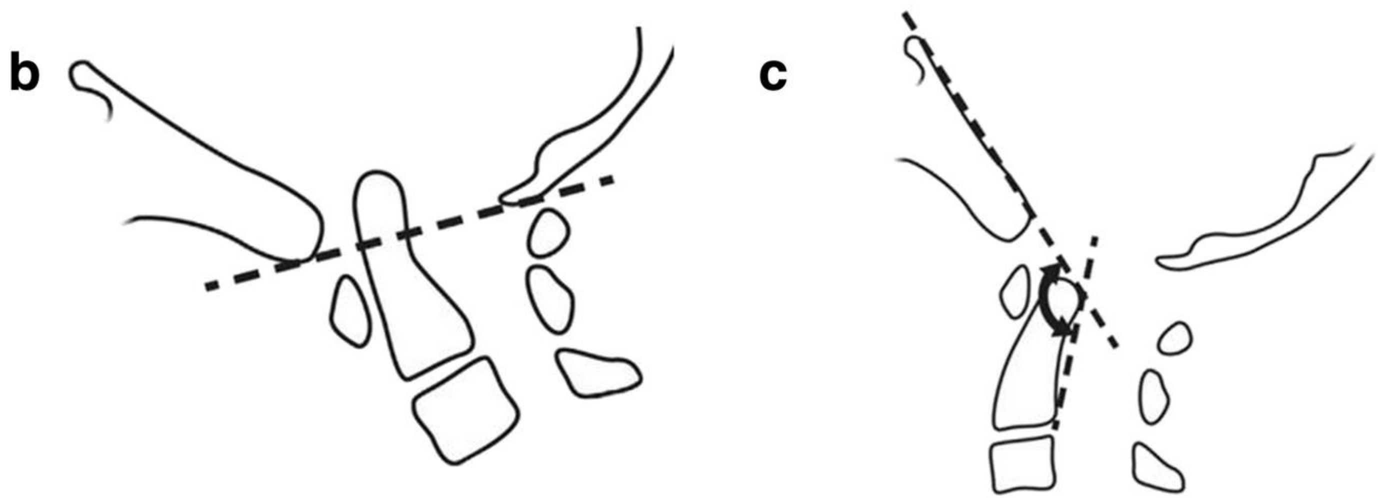
1. Clivoaxial angle <125
2. BI
3. Medullary Kinking
4. Chiari 1.5
5. pB-C2

Ho et al., 2019

pB-C2 >9mm–
ventral canal encroachment



***Flex/Ex Xray/MRI helpful**



McRae’s Line 5mm below –
Basilar Invagination

Clivo-axial angle <125–
Assoc w/ platybasia

Condular-C2 Sagittal Vertical Alignment (C-C2SVA >= 5mm)

JNS PEDIATRICS

CLINICAL ARTICLE

J Neurosurg Pediatr 28:176–182, 2021

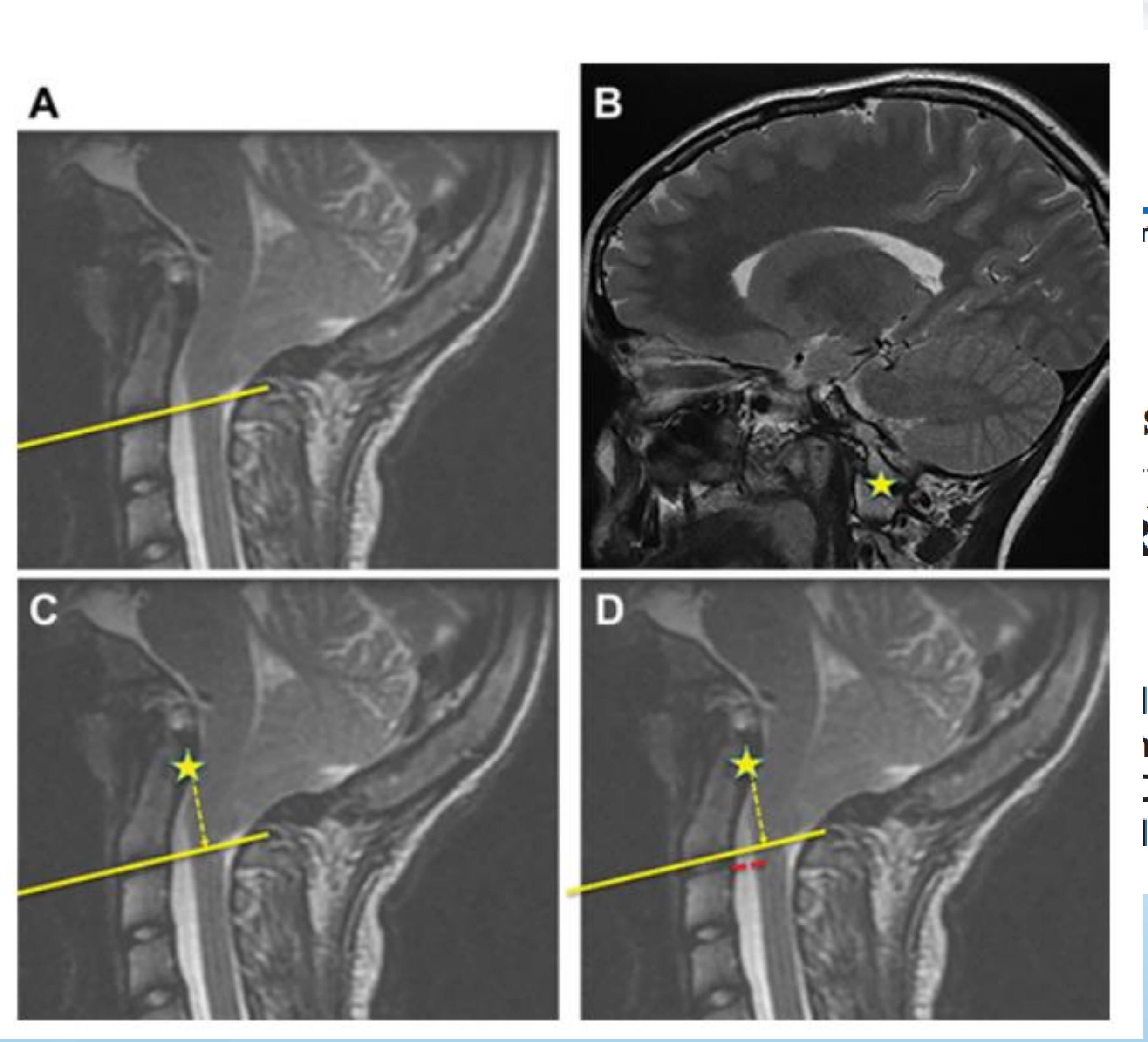
A multicenter validation of the condylar–C2 sagittal vertical alignment in Chiari malformation type I: a study using the Park-Reeves Syringomyelia Research Consortium

Vijay M. Ravindra, MD, MSPH,[1–3] Rajiv R. Iyer, MD,[1] Alexander T. Yahanda, MS,[4] Robert J. Bollo, MD,[1] Huirong Zhu, PhD,[5] Evan Joyce, MD, MS,[1] Tammy Bethel-Anderson,[4] Thanda Meehan, RN, BSN,[4] Matthew D. Smyth, MD,[4] Jennifer M. Strahle, MD,[4] Tae Sung Park, MD,[4] David D. Limbrick Jr., MD, PhD,[4] and Douglas L. Brockmeyer, MD,[1] on behalf of the Park-Reeves Syringomyelia Research Consortium

Condular-C2 Sagittal Vertical Alignment (C-C2SVA >= 5mm)

JNS PE

CLINICAL ARTICLE
rosurg Pediatr 28:176–182, 2021



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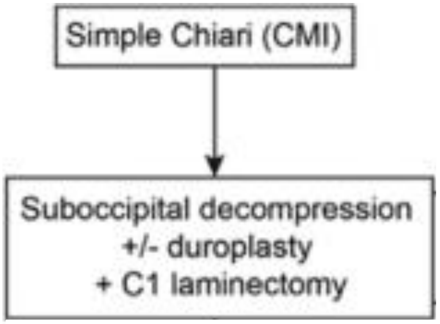
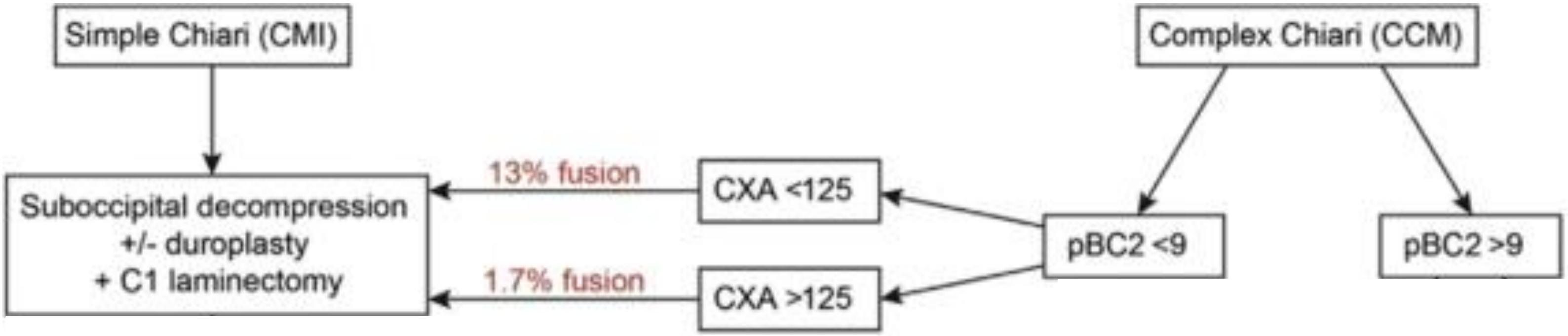

Simple Chiari (CMI)
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy

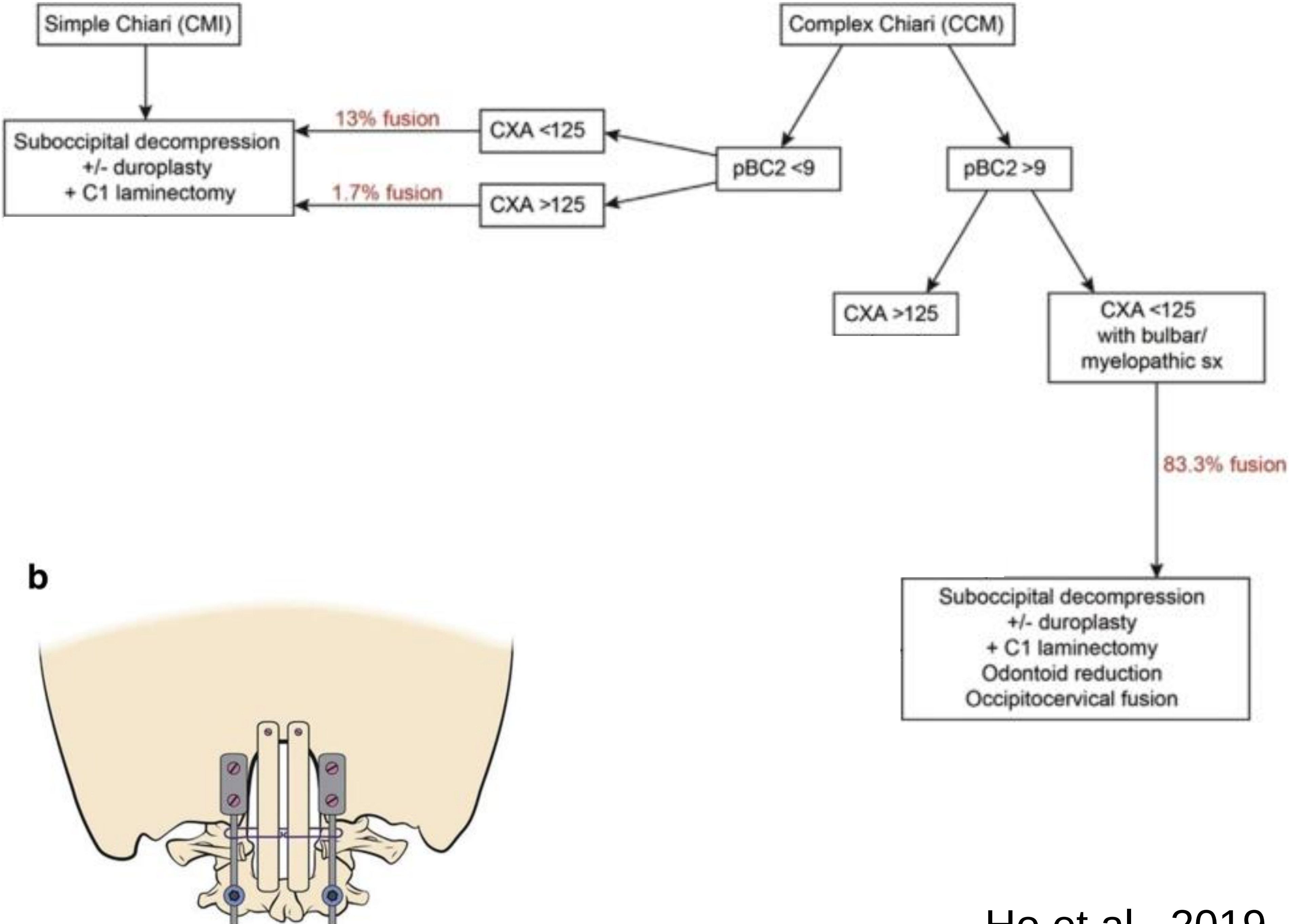
Ho et al., 2019

Simple Chiari (CMI)
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
Complex Chiari (CCM)
pBC2 <9
pBC2 >9

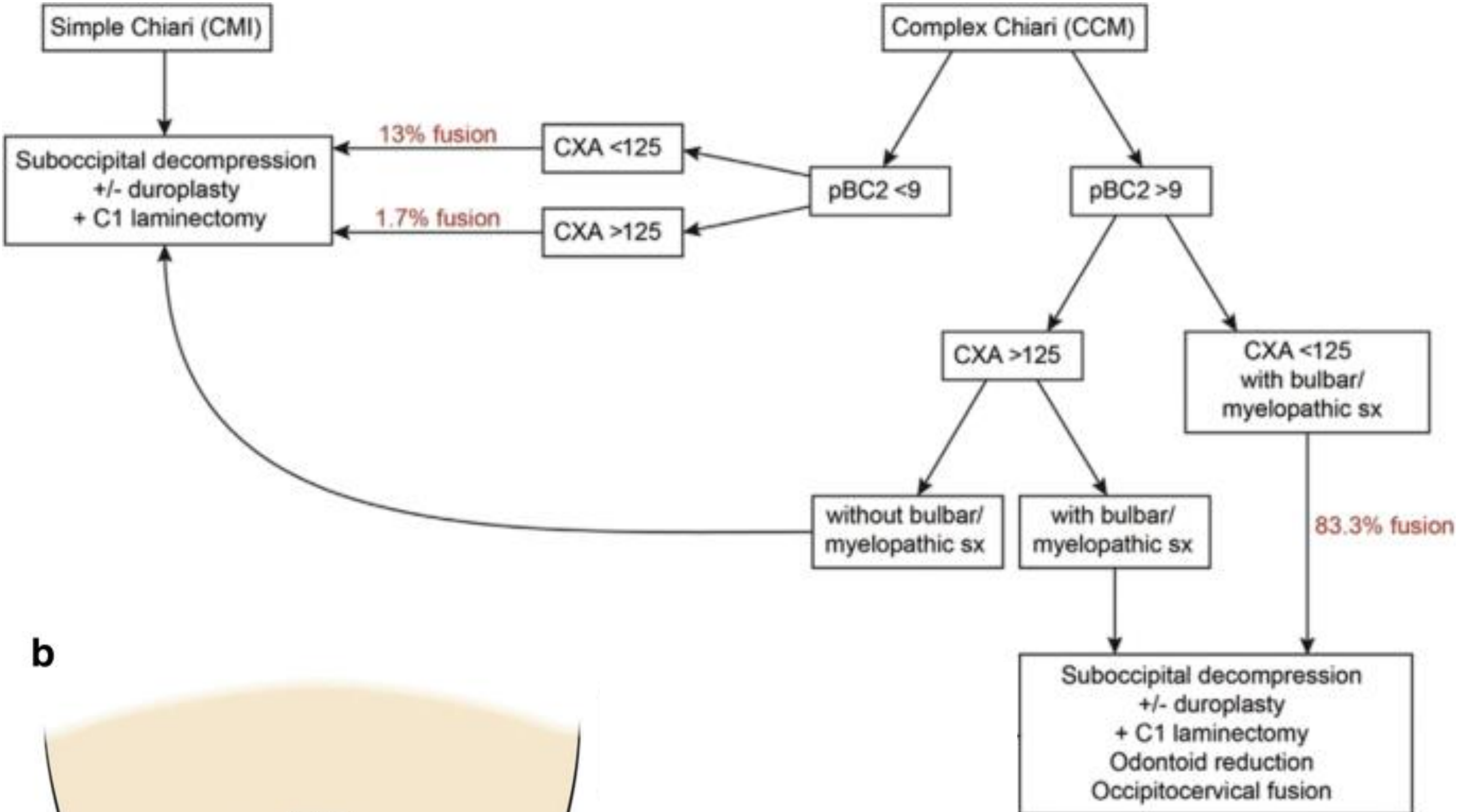
Ho et al., 2019


Simple Chiari (CMI)
Complex Chiari (CCM)
13% fusion
CXA <125
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
1.7% fusion
CXA >125
pBC2 <9
pBC2 >9

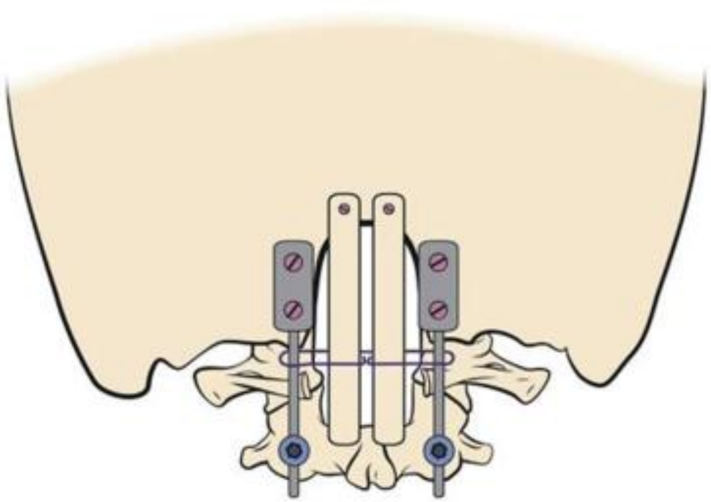
Ho et al., 2019


Simple Chiari (CMI)
Complex Chiari (CCM)
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
13% fusion
CXA <125
1.7% fusion
CXA >125
pBC2 <9
pBC2 >9
CXA >125
CXA <125
with bulbar/
myelopathic sx
83.3% fusion
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
Odontoid reduction
Occipitocervical fusion
b

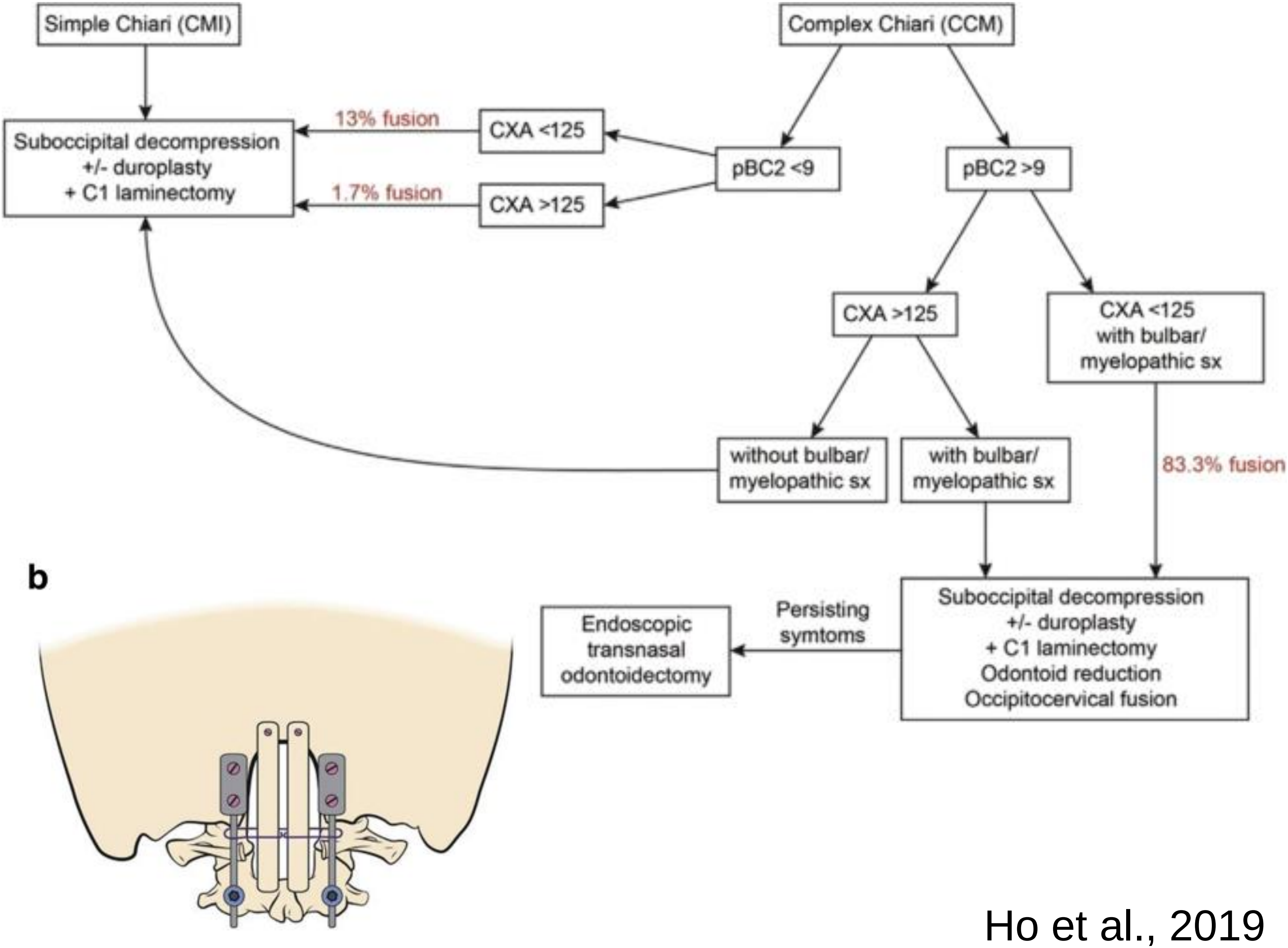
Ho et al., 2019



b



Ho et al., 2019



Simple Chiari (CMI)
Complex Chiari (CCM)
13% fusion
CXA <125
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
1.7% fusion
CXA >125
pBC2 <9
pBC2 >9
CXA >125
CXA <125
with bulbar/
myelopathic sx
without bulbar/
myelopathic sx
with bulbar/
myelopathic sx
83.3% fusion
b
Endoscopic
transnasal
odontoidectomy
Persisting
symtoms
Suboccipital decompression
+/- duroplasty
+ C1 laminectomy
Odontoid reduction
Occipitocervical fusion

Ho et al., 2019

Objectives

1. Describe the definition and classification of “Chiari Malformations”

2. Describe Syringomyelia

3. Explain the association of Chiari I malformation and Syringomyelia and the pathophysiological theories explaining this

4. Choose appropriate therapy of Chiari I malformation with or without syringomyelia